

ONTARIO
SUPERIOR COURT OF JUSTICE

BETWEEN:)
)
The Estate of Carlo DeMarco)
) J. Makins and J. Nicholson, for the Plaintiffs
Plaintiffs)
)
– and –)
)
Dr. Rene Frances Martin) S. Rogers and J. Sirivar, for the Defendant,
) Dr. Martin
Defendants)
)
)
)
) **HEARD:** September 10-12, 14, 18, 19 and
) 25-28 and October 2, 3, 12 and 15, 2018

MITCHELL J.

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INTRODUCTION

- [1] On August 21, 2011, Carlo DeMarco died suddenly. He was 51 years old. An autopsy revealed he suffered from advanced coronary artery disease.
- [2] On February 10, 2011, Mr. DeMarco's family physician, the defendant, Dr. Rene Frances Martin, performed a physical examination of Mr. DeMarco. Dr. Martin noted no concerns with Mr. DeMarco's health and, in particular, noted no symptoms of coronary artery disease.
- [3] On June 13, 2011, Mr. DeMarco attended an appointment with Dr. Martin complaining of indigestion, left arm discomfort and feeling faint. Dr. Martin prepared a letter of referral to Dr. Dennis Wong, a cardiologist, enquiring about conducting a stress test. The referral was not sent.

OVERVIEW OF THE CLAIM

- [4] The plaintiffs in this action are comprised of the estate of Mr. DeMarco, Mr. DeMarco's spouse, Elizabeth (Lisa), his four adult children – Emily, Gabriel, Franco and Julia, and his parents, Franco and Rita.
- [5] In this action, the plaintiffs allege that Dr. Martin failed to make enquiries of Mr. DeMarco during the physical examination conducted on February 10, 2011 sufficient to elicit the symptomology of which Mr. DeMarco later complained in June 2011, in breach of the standard of care. The plaintiffs further allege that Dr. Martin breached the standard of care by failing to send the stress-test referral to Dr. Wong on June 13, 2011.
- [6] Dr. Martin denies he breached the standard of care by failing to elicit the reporting of symptoms by Mr. DeMarco at the February 2011 periodic examination. However, Dr. Martin admits that his failure to send the referral to Dr. Wong on June 13, 2011 constitutes a breach of the standard of care.
- [7] Causation is the central issue in this case. In his defence, Dr. Martin submits that even had he sent the referral to Dr. Wong on June 13, 2011, Mr. DeMarco's coronary artery disease would not have been diagnosed and treated prior to the events of August 20, 2011 which led to his sudden death (the "wait-time defence"). To support his position, he relies on Dr. Wong's average wait time for conducting a stress test in the Summer of 2011 and Dr. Jablonsky's average wait time to conduct an angiogram and ultimately perform revascularization through stenting or surgery in the Summer of 2011.
- [8] Dr. Martin also defends on the basis that had Mr. DeMarco's coronary artery disease been detected and treated using medical management and/or revascularization (bypass or angioplasty) prior to August 20, 2011, Mr. DeMarco's life would not have been

prolonged because the arrhythmia which led to his death was caused by scarring on his heart muscle and not complications arising from his untreated advanced coronary heart disease (the “treatment defence”).

- [9] The damage claims of the *Family Law Act* (“FLA”) claimants have been agreed by the parties subject to the court’s approval of the settlement relating to the FLA claim of Emily, a party under a disability. However, certain claims for past and future special damages and the economic loss claims (dependency loss claims) of Lisa and Emily have not been resolved.
- [10] Before the trial commenced, the action was dismissed against all defendants except Dr. Martin.

THE EVIDENCE

A. Personal History of Carlo DeMarco

- [11] At the time of his death on August 21, 2011, Mr. DeMarco was married and the father of four adult children. He was self-employed in the construction industry designing and building new homes and performing residential and commercial renovations. He owned and operated a company carrying on business as “Carriage Hill Construction” (“CHC”).
- [12] Mr. Demarco had 4 years of university education in drafting/design when he started CHC in London, Ontario in 1993. Both of his sons were employed by CHC as general labourers. In 1997, Lisa left her fulltime employment with Bell Canada to become involved in CHC. Upon leaving Bell, she worked full-time performing administrative tasks for CHC as well as caring for the home and Emily.
- [13] In June 2011, the family home was located in an exclusive enclave of homes in London. The family home was a high-end home designed and built by Mr. Demarco on a lot purchased in 1997. At the time of Mr. Demarco’s death, all four children resided in the home; however, only Lisa, Emily and Gabriel were at home when Mr. DeMarco went into medical distress on August 20, 2011.

B. Carlo DeMarco’s Medical History

- [14] Mr. DeMarco was a patient of Dr. Martin since 2007 when his previous family physician, Dr. Gambrill, retired. Dr. Martin first met with Mr. DeMarco on January 29, 2007 as a new patient. They discussed Mr. DeMarco’s ankle injuries suffered as a teenager and his carpal tunnel syndrome. At this meeting, Dr. Martin referred Mr. DeMarco to an orthopedic surgeon with respect to his ankle issues and to a plastic surgeon with respect to his carpal tunnel syndrome.
- [15] Mr. DeMarco attended at Dr. Martin’s offices for advice and treatment on numerous occasions during the period 2007 through 2011 as follows:

- (a) he attended on January 11, 2007 as a new patient;

- (b) he attended a periodic physical examination on December 11, 2007;
- (c) he attended on April 28 and May 15, 2008 with respect to a mole;
- (d) he attended on July 8, 2008 with respect to chronic low-back pain;
- (e) on December 28, 2010 he attended to report right hand pain;
- (f) he attended a periodic physical examination on February 10, 2011;
- (g) on May 31, 2011, he attended and was treated for a wart on his finger; and
- (h) he last attended on June 13, 2011 to report indigestion, left arm discomfort and feeling lightheaded.

[16] His medical records reveal he was diagnosed and treated for carpal tunnel syndrome and ankle injuries long before becoming a patient of Dr. Martin. It is not disputed that Mr. DeMarco's cardiac risk factors were low. At the time of his death at the age of 51 years, Mr. Demarco was a non-smoker and a minimal drinker. He did not suffer from diabetes or hypertension. He did not have high cholesterol. He was not overweight. He had no family history of heart disease. He did not take prescription medications. He exercised regularly by walking and jogging once a month. Lisa testified that she and Carlo took regular long walks together and "she could not keep up with him". When exercising or otherwise exerting himself, Mr. Demarco reported no symptoms of angina – pain, pressure or tightness in his chest.

C. Carlo DeMarco attends the ER on July 24, 2010

- [17] On July 24, 2010, Mr. Demarco fainted and attended at the hospital with Lisa. He was seen by a cardiologist and discharged later that same day. It was concluded by the attending cardiologist that the fainting episode was unrelated to any cardiac condition rather was vasovagal in nature.
- [18] Mr. DeMarco did not report the July 24th fainting episode when he attended with Dr. Martin in December 2010, or at his February and May 2011 attendances. It was not until June 13, 2011 that Dr. Martin was first advised of this episode of syncope.

D. Dr. Martin's Practice – Periodic Health Review

- [19] Mr. DeMarco attended on December 11, 2007 for his first of two periodic physical examinations conducted by Dr. Martin. Prior to the examination, Dr. Martin had reviewed Mr. DeMarco's medical records provided by his former family physician. Dr. Martin described using a template to complete his electronic medical record relating to his physical examination of Mr. DeMarco with "N" denoting "negative" or an absence of symptoms.
- [20] With respect to his usual approach to conducting a physical examination of a patient, Dr. Martin gave the following evidence:

and Q. So starting first with the review of systems, what is a review of systems what is it that you're doing?

A. So when someone comes in for their check-up we ask, how are you feeling? Well, not well. If they are not well, then I spend a lot of time on the not well part of the story. If they feel well then I want to make sure that they really are well. Because some people say they are well but are really not or forget to tell me something, etc.

...

Q. So in a patient like Mr. DeMarco –

A. For most people who are well it's usually just a head-to-toe thing because it's easy.

Q. And so going head to toe, what are the nature of the questions you would have asked Mr. DeMarco in the system that you reviewed in order to determine that the review of system was negative?

A. So when someone feels well it's a head to toe thing. So it's things like any headaches? How is your vision? Any hearing problems? You go see do you they see the dentist. Then in the chest, any shortness of breath? Now, if smoke cigarettes I will spend a bit more time on that.

Q. And in this case he told you he didn't smoke cigarettes?

A. So if they don't smoke it would just be something quick. No shortness of breath, no coughing, no chest pain. No, everything is good. Then I move down the abdomen. Bowels are working okay. Any heartburn, any tummy upset? Keep going. The next system in men it's any pain in the groin, any swelling in the groin, any erection problems after a certain age. Then down your legs to legs. So that's when I ask just in general any aches and pains in or arms and musculoskeletal problems.

And at the end of that I ask about how is their mood? How do you handle stress? I've already asked them about smoking and alcohol, that is usually the starting.

So then I have all those and if they say yes to any of those things there is more questions and more questions. But if they are all no then I say, okay, I'm going to leave the room and cover up with the gown. I'll be back in two minutes.

Then I go back and begin my physical examination, which again is pretty orderly, from top down.

...

- A. So review of systems is done mainly before the check up and then added onto during the physical checkup. So I'm talking while I'm examining, asking these questions.

E. Periodic Health Review Conducted February 10, 2011

- [21] In 2011, Dr. Martin had been a practising full-time family physician for over 22 years. He was familiar with the symptoms of coronary artery disease. He described angina as the classic symptom of coronary artery disease characterized by "the recurrence of chest discomfort with exertion".
- [22] Dr. Martin's notes prepared contemporaneous with the examination conducted on February 10, 2011 do not report any of the symptoms complained of by Mr. Demarco four months later on June 13, 2011, namely, indigestion (dyspepsia), left arm discomfort and feeling faint. Specifically, his notes do not contain any reference to the fainting spell and attendance in the ER on July 24, 2010.
- [23] At the February 10, 2011 attendance, Mr. Demarco did not complain of symptoms consistent with coronary artery disease. Notes recorded by Dr. Martin on February 10, 2011 indicate as follows:

BP: 132/82 Wt: 211

Annual: Generally feels well, except recurrent back strain. ROS neg.

O/E:

Nodes: N

Ent: N

Thyroid: N

Chest: N

Abd: N

Rectal: N

Testes: N

CNS: N Skin: N

Dx: Annual

P:1 BW, pros/cons PSA, FOBT 2. Rx for back. He mentioned wt loss helped in the past. Agree, maybe not down to 170 lbs like he did a couple years ago.

- [24] When asked what questions would have been asked of Mr. DeMarco in order to elicit the information to support the entry in the medical record stating: "Generally feels well except recurrent back strain", Dr. Martin answered:

even Well, same thing. So now I've met him a few times and other than, you know, back in December his only stated concern that day was his right hand.

So, again, how are you doing? I'm well today. Do you feel well generally? Do you feel well most days? Some kind of expression like that. Yes. And he would have told me he has some back strain from time to time. Fine, and the questions there would be, doesn't slow you down very much? Have you ever had treatment for your back? Do you have to take Advil frequently for your back? Have you gone for physiotherapy for your back? Whatever. So those would be the types of questions for his back pain.

- [25] Dr. Martin's notes indicate his review of body systems as "negative" which he described as meaning that Mr. DeMarco did not have any major complaint in any of the major systems of the body. At the end of the examination, Dr. Martin ordered blood work to test Mr. DeMarco's sugar and cholesterol levels and he provided advice for the treatment of his back pain. He also suggested weight loss. The results of blood tests conducted in March 2011 were normal.
- [26] Dr. Martin testified that it was not his practice to enquire about existing or chronic conditions unless they were worsening. In the case of Mr. DeMarco, he has no specific recollection of the physical examination conducted on February 10, 2011 but would have followed his usual practice. As was his usual practice, he would not have enquired about Mr. DeMarco's carpal tunnel syndrome or his vasovagal episodes because, in his view, both were chronic conditions and not noteworthy unless the conditions were worsening. If Mr. DeMarco had mentioned any specific concerns or symptoms he would have made a record of those concerns as was his practice, in the same manner as he did for Mr. DeMarco's concern regarding his back pain.

F. Examination Conducted June 13, 2011

- [27] Notes recorded by Dr. Martin on June 13, 2011 indicate as follows:

Fainted last year. See in ER. Since then, intermittent dyspepsia, followed by persyncope (sic), occ left arm discomfort. No CP or palpitations Doesn't feel anxious at the time

BP 120/82 P=56 reg

Abd n HS n no murmurs

A+P Symptoms NYD>Stress, Cardiac risk factor low, but will arrange a stress test to be sure.

- [28] On June 13, 2011, Mr. DeMarco reported to Dr. Martin that since July 2010 he had been experiencing intermittent dyspepsia, followed by presyncope and occasional left arm discomfort (the "triad of symptoms"). Dr. Martin testified that he did not believe Mr. DeMarco was suffering from angina because he did not report any of the classic symptoms of angina being chest pain or discomfort upon exertion. Mr. DeMarco did not report heart palpitations. Dr. Martin knew Mr. DeMarco was active and worked full-time in the construction industry.

- [29] Dr. Martin specifically recalls the look on Mr. DeMarco's face on June 13, 2011 and remembers thinking that "there was something else going on that Mr. DeMarco was unable to articulate". Not being sure what that might be, Dr. Martin indicated in his notes "symptoms NYD" (not yet diagnosed). He made a referral for a stress test so as to rule out a cardiac cause for the triad of symptoms. He knew that a cardiac-related cause of the triad of symptoms was potentially life threatening.
- [30] Following his June 13th attendance and during the time leading to his death, a little more than two months later, Mr. DeMarco did not follow up with Dr. Martin with respect to the referral, he did not tell family and specifically Lisa that he had attended an appointment with Dr. Martin on June 13, 2011 or that Dr. Martin had made a referral to Dr. Wong for a stress test.
- [31] Lisa testified that she was unaware Mr. DeMarco had made an appointment and attended with Dr. Martin on June 13, 2011. Mr. DeMarco did not tell Lisa he had been experiencing the triad of symptoms. She did recall that approximately two days prior to his death he complained of feeling faint and was nauseous. He also mentioned numbness in his wrist which Lisa assumed related to his carpal tunnel syndrome. Mr. DeMarco did not tell Lisa about his left arm discomfort.
- [32] Lisa suggested to Mr. DeMarco that he call his doctor but was not concerned when he did not because there was no reason to believe he was suffering from any cardiac related condition – he had no history of and was not experiencing any typical symptoms of coronary artery disease. Lisa acknowledged that Mr. DeMarco was the kind of person who did not delay going to a doctor if he had concerns with his health. Mr. DeMarco attended his medical appointments on his own, unaccompanied by Lisa.

G. Events of August 20, 2011

- [33] After visiting a relative in hospital earlier that evening, Lisa and Mr. DeMarco returned home at approximately 9:30 p.m. on August 20, 2011. After eating dinner, Lisa and Mr. DeMarco spent time together relaxing and Lisa left Carlo in bed at approximately 11:15 p.m. and went across the hall to watch TV. Shortly thereafter, Lisa heard gasping coming from the bedroom. She went to the bedroom and found Mr. DeMarco turning grey and gasping for breath. Gabriel, a trained firefighter, was called from the basement and began performing CPR on his father.
- [34] EMS arrived at approximately 11:30 p.m. EMS "shocked" Mr. DeMarco's heart seven times in an effort to resuscitate him. These efforts were not successful. Mr. DeMarco was pronounced dead at London Health Sciences Centre at 12:28 a.m. on August 21, 2011.
- [35] Shortly after learning of Mr. DeMarco's death and upon reviewing his records, Dr. Martin summonsed Lisa and Mr. DeMarco's children to his office and advised them of Mr. DeMarco's attendance on June 13, 2011. He admitted to inadvertently failing to send the stress test referral to Dr. Wong.

H. Autopsy Findings

- [36] Dr. Subrata Chakrabarti, a pathologist with London Health Sciences Centre, performed an autopsy on Mr. Demarco's body approximately 12 hours following his death. Dr. Chakrabarti did not testify, rather, portions of his report were admitted for their truth as evidence.¹ It was admitted by the parties as follows:

Pre-Autopsy Information

51 year old man with sudden death, clinical diagnosis of myocardial infarction/coronary artery disease.

Cardiovascular System

Heart Weight: 440 g

Coronary Arteries:

- LAD – up to 90% stenosed
- RCA - up to 90% stenosed
- Circumflex – 50% stenosed

Atria: unremarkable

Cardiac Valves: TV=14.0 cm PV=8.0 cm MV=11.0 cm AV = 7.5 cm

Right Ventricle: Wall thickness, 0.6 cm

Left Ventricle: wall thickness. 2.0 cm. 2 paler areas; 1) posterolateral; 2) posteroseptal

Aorta & Major Arteries: unremarkable

Vena Cavae & Major Veins: unremarkable

Microscopic Description

Heart: patchy fibrosis in both ventricles, myocytic hypertrophy and moderate to severe coronary atherosclerosis with stenosis up to 90% in left anterior descending to right coronary arteries; circumflex 50% stenosis; there is no evidence of acute myocardial infarction; possible focus of contraction band necrosis present

Lungs: marked congestion and edema

¹ See underlined portions of Exhibit No. 17.

Major Findings

- 1) Severe atherosclerosis of left anterior and right coronary artery, with stenosis up to 90%.
- 2) Numerous patchy microscopic old healed scars, contraction band.
- 3) Moderately congested and edematous lungs (right 737 grams, left 527 grams).
- 4) Bilateral cortical benign cysts (kidneys).
- 5) Liver, moderate congestion.
- 6) Stomach, chronic gastritis, nonspecific.

I. Wait times for stress testing and treatment of coronary artery disease

(i) Dr. Wong's estimated wait time for a non-urgent stress test referral in June 2011

- [37] Dr. Dennis Wong is a general internist practicing cardiology with a group of cardiologists at the London Cardiac Institute, an outpatient clinic. He is the cardiologist to whom Dr. Martin referred his patients in June 2011 for a stress test. Dr. Martin admits that had the referral been sent to Dr. Wong, he would have received a response from Dr. Wong's office with an appointment date before the events of August 20, 2011 took place leading to Mr. DeMarco's death. Dr. Wong corroborated this evidence.
- [38] An issue was raised with regards to the manner in which Dr. Martin prepared the referral. The referral did not definitively recommend a stress test rather left that decision open to Dr. Wong. Dr. Wong testified that regardless of the language used by Dr. Martin, his practice is to defer to the referring physician and conduct a stress test if recommended or suggested. He said that in only the rarest of cases would he decline to order a stress test.
- [39] Dr. Wong testified that he would independently assess whether the stress test would be scheduled on a non-urgent or urgent basis. Dr. Wong testified that, based on Mr. DeMarco's unremarkable medical history with respect to as his low probability for coronary artery disease and his review of Dr. Martin's referral note, he would have scheduled Mr. DeMarco for a non-urgent stress test.
- [40] Before conducting a stress test, Dr. Wong conducts an assessment of a patient's "pre-test probability" of having heart disease. A patient referred to Dr. Wong having the cardiac history of Mr. DeMarco would have been assessed as having a low pretest probability for heart disease. Once the results of the stress test are known (abnormal or normal), they are considered together with the pre-test probability to determine whether the patient likely has heart disease.
- [41] Dr. Wong testified that in his experience a patient with significant two-vessel coronary artery disease, as was the case with Mr. DeMarco, would likely have an abnormal stress

test. Dr. Wong testified that in a young patient with low pre-test probability for heart disease and an abnormal stress test he would likely refer that patient on for further testing – a nuclear stress test or an angiogram. A nuclear stress test would be conducted by the hospital and the patient referred back to Dr. Wong to review the results approximately 3-4 weeks later.

- [42] Dr. Wong referred all patients for whom he recommended an angiogram to Dr. George Jablonsky. Dr. Wong admitted that a patient referred for an angiogram would be started on medication, typically Aspirin, and a medication to treat angina and reduce its symptoms (beta blockers). Dr. Wong testified that other than helping to prevent a heart attack, Aspirin would not assist in treating coronary artery disease such as a plaque rupture.
- [43] By correspondence dated August 20, 2015 addressed to defendant's counsel, Dr. Wong estimated that his average wait time in the summer of 2011 for a stress test for patients referred to him on a non-urgent basis was 6-8 weeks. He referred to a sample of 10 or less charts of patients referred to him in June 2011 and determined that the average wait-time was closer to the 8-week mark.² In correspondence dated September 3, 2015 addressed to defendant's counsel, Dr. Wong provided two specific examples – a referral received June 14, 2011 and a referral received June 15, 2011 were both seen on August 17, 2011.
- [44] As trial approached, Dr. Wong further refined his estimate to an average wait time of 59 days. In providing this further estimate, Dr. Wong relied on a summary of all (75 in total) records for patients referred to him in June 2011 on a non-urgent basis. This summary chart was prepared by the office administrator of his clinic. He described the process used to arrive at his 59-day estimate. First, software code was written to extract only those patient records relating to patients referred for a non-urgent stress test in June 2011 from all electronically-stored patient records (approximately 700-800 patient charts). Next, a technologist in Dr. Wong's offices reviewed the 75 patient charts and summarized the pertinent information necessary to calculate the average wait time. Last, Dr. Wong personally compared 10 of these 75 patient charts against the information contained in the summary chart to verify its accuracy.
- [45] The plaintiffs challenged the admissibility of Dr. Wong's evidence on the basis it constituted hearsay. I found that Dr. Wong's evidence was not hearsay and was admissible in its entirety. I held that the concerns raised with respect to the chart's reliability related to the weight to be ascribed to Dr. Wong's evidence.³

(ii) Actual Wait Times of Lisa DeMarco for appointment with Dr. Wong

- [46] Lisa had also been referred to Dr. Wong by her family physician in the past for cardiac-related issues. The plaintiffs rely on Lisa's specific wait times as evidence in support of their position that the average wait time for a referral to Dr. Wong was far less than 6-8

² See Exhibit Nos. 39 and 41.

³ Reasons reported at *The Estate of DeMarco et al. v. Martin*, 2018 ONSC 5948.

weeks. Lisa's evidence as to her wait times for an appointment with Dr. Wong was as follows:

- (a) Lisa was given a non-urgent referral by her family physician to Dr. Wong on January 16, 2008. She was seen by Dr. Wong and testing was performed on January 31, 2008 – 15 days later. She had a follow up appointment on February 19, 2008 and a further follow-up appointment on March 27, 2008;
- (b) Lisa was referred to Dr. Wong on November 2, 2011. She was originally given an appointment date of January 3, 2012 (62 days later) however requested to be put on the cancellation list and saw Dr. Wong on November 8, 2011 – 6 days later; and
- (c) Lisa was referred to Dr. Wong on April 10, 2013 and saw him on May 6, 2013 – 26 days later.

[47] During cross-examination Lisa admitted that heart abnormalities were found during an electrocardiogram in late 2007 (cardiomyopathy) and that she had a different heart history than Mr. DeMarco. She admitted to having high cholesterol and a family history of heart disease with her father, uncles and grandfather dying at a relatively young age from heart disease.

(iii) Wait time for angiogram/angioplasty with Dr. Jablonsky

[48] If an angiogram is recommended, Dr. Wong refers his patients to Dr. Jablonsky.

[49] Dr. George Jablonsky was asked to assume that Mr. DeMarco would have been seen on the earlier of the 6-8 week average wait-time spectrum for a stress test referral in the summer of 2011. He was asked to further assume the stress test took place on July 25, 2011, that the stress test was abnormal during the later stages, and that Dr. Wong made an immediate referral to Dr. Jablonsky. Dr. Jablonsky testified that he would have seen Mr. DeMarco at his first clinic upon his return from vacation (July 22- August 22, 2011) on August 24th, 2011.

[50] On cross-examination, Dr. Jablonsky estimated that had Mr. DeMarco been triaged following his stress-test he would have been referred to a more generic cardiac group in Dr. Jablonsky's absence. He testified that had he been triaged, Mr. DeMarco's wait time would have been reduced from 4 to 3 weeks.

[51] Dr. Jablonsky explained that the longer the narrowing, the more tortuous the artery and the more side branches, the less likely an angioplasty would be performed. The more metal you put in the artery, the higher the chance of stenosis and thrombosis of the stent. These are factors considered when a patient has double-vessel disease as to whether bypass surgery or angioplasty is more suitable.

[52] Dr. Jablonsky was of the view that had bypass surgery been an option for Mr. DeMarco, the likelihood he would have had revascularization surgery on an urgent basis would

have been low. Dr. Jablonsky estimated a wait-time of one to two months for bypass surgery.

J. Pre-test Probability and Likely Outcome of Stress Test

- [53] After reviewing the notes of Dr. Martin from Mr. DeMarco's June 13, 2011 attendance, Dr. Jablonsky was of the opinion that DeMarco was not suffering from angina because he did not report any of the typical symptoms. For example, Mr. DeMarco did not have chest pain on exertion. He complained of indigestion but no exertional discomfort compatible with angina. He did not complain to his family of any discomfort. He worked long hours. He did not report to his physician exertional chest discomfort. His blood pressure was normal at 120/80. His heart rate was normal with a pulse of 56. His heart sounds were normal.
- [54] Dr. Jablonsky was of the view that Mr. DeMarco's cardiac risk factors were "very low to nonexistent."
- [55] Drs. Wong and Jablonsky indicated Mr. DeMarco's pre-test probability would have been "very low". None of the plaintiffs' experts disputed this evidence. In the absence of chest pain, Dr. Jablonsky did not consider left arm discomfort as being indicative of cardiac disease.
- [56] Dr. Jablonsky was of the opinion that the referring cardiologist would have ordered a stress test upon receipt of the referral despite the absence of risk factors. He explained that there is a lot of benefit or value to a patient in a negative test.
- [57] Having regard to the pathological results on autopsy, Dr. Jablonsky believed that the likely outcome of Mr. DeMarco's stress test would have been abnormal in the later stages of the test.
- [58] Dr. Fitchett agreed with Drs. Wong and Jablonsky that Mr. DeMarco's pretest probability would have been low. He further agreed with Drs. Wong and Jablonsky that the stress test would have been abnormal but at a moderate exercise load.
- [59] All experts agreed that at some stage of the stress test, Mr. DeMarco's coronary artery disease would have revealed itself. That is, the stress test would have produced abnormal results. However, the experts were not in agreement with the stage at which abnormal results would have been generated. Drs. Jablonsky and Wong were of the opinion that it would have been late stage abnormal results and Drs. Fitchett and Myers were of the opinion that it would have been mid-stage abnormal results.

ISSUES

- [60] There are three issues to be decided in this action. They are:

- (i) Did the defendant breach the standard of care of a family physician in conducting Carlo DeMarco's periodic health examination on February 10, 2011?
- (ii) Did any breach of the standard of care by Dr. Martin on February 10, 2011 and/or June 13, 2011 cause Carlo Demarco's death? More specifically: (a) would Carlo DeMarco have been diagnosed with and treated for coronary artery disease prior to August 20, 2011?; and (b) did Carlo DeMarco's coronary artery disease cause the fatal arrhythmia he suffered on August 20, 2011?
- (iii) What is the appropriate measure of damages?

RELEVANT LEGAL PRINCIPLES

A. General

- [61] The plaintiffs must establish a breach of the standard of care, that the breach caused the plaintiffs' injury or loss and the quantum of such loss, on a balance of probabilities.
- [62] In a medical malpractice action, expert evidence is necessary to assess both standard of care and causation. In assessing the conduct of a physician, the court "must be cautious to base their conclusions upon the expert evidence before them, and not to speculate as to the adequacy of professional standards in the absence of expert evidence attacking those standards."⁴

B. Standard of Care

- [63] Every medical practitioner must bring to his task a reasonable degree of skill and knowledge and must exercise a reasonable degree of care. He is bound to exercise that degree of care and skill which could reasonably be expected of a normal, prudent practitioner of the same experience and standing.⁵
- [64] The standard of care at the time of the alleged negligence is the relevant standard of care, not the standard of care at the time of trial. The medical professional's conduct must be

⁴ *Hajgato v. London Health Association*, [1982] O.J. No. 2564 (H.C.), at para 36; aff'd (1983), 44 O.R. (2d) 264 (C.A.).

⁵ See *Crits v. Sylvester* (1956), O.R. 132 (C.A.) at 143; aff'd [1956] S.C.R. 99; *Milne v. St. Joseph's Health Centre*, 2009 CanLII 51196 (Ont. Sup.Ct.) at para. 69.

judged in light of the knowledge that existed at the time of the alleged acts of negligence, not knowledge subsequently learned.⁶

[65] A lack of charting does not necessarily mean that procedures were not conducted nor is the mere lack of charting *prima facie* evidence of negligence in conducting the examination. However, the lack of charting makes it more difficult for a court to determine matters of credibility where individuals who are trained to chart did not do so. This failing, despite the opportunity to do so, makes it more difficult for a court to accept that the correct steps were followed and questions asked as it would have been logical for them to be recorded had they been done.⁷

[66] The standard of care is a standard of reasonableness. What a physician would do themselves in a particular situation does not establish the standard of care. As stated in *Bafaro v. Dowd*:

To the extent that an expert testifies as to what he himself would do in a situation, rather than what the standard of care requires, his testimony does not establish the standard of care nor demonstrate that the defendant doctor breached a standard of care.⁸

[67] The usual practice of a physician can provide circumstantial evidence as to what occurred at a patient encounter, if the physician has no memory of the encounter. However, like all circumstantial evidence, the value of the evidence of usual practice lies in the inferences that can reasonably be drawn from it. The inferences that can be drawn depend on the regularity of the practice and all of the other evidence in the case, particularly other direct or circumstantial evidence that impacts on whether the practice was followed on the occasion in question.⁹

C. Causation

[68] The test of causation is the “but for” test. The “but for” test is a factual inquiry, and applies to single and multi-cause injuries. The plaintiffs must demonstrate, on a balance of probabilities, that “but for” the defendant’s conduct, the plaintiffs would not have suffered the injury or loss. In other words, the defendant’s negligence was necessary to bring about the harm.¹⁰

[69] Where the “but for” test is unworkable, courts have adopted a material contribution test to determine causation. This test is an exception to the “but for” test and is only to be used in special circumstances as follows:¹¹

⁶ *Grass (Litigation guardian of) v. Women's College Hospital*, [2003] O.J. No. 5313 (Sup. Ct.) at para. 170.

⁷ *Ghiassi v. Singh*, 2017 ONSC 6541, para. 25, aff’d 2018 ONCA 764.

⁸ [2008] O.J. No. 3474 (S.C.J.) at para. 36, aff’d [2010] O.J. No. 979 (C.A.).

⁹ *Taylor v. Morrison*, 2006 CanLII 24463 (ON SC), paras. 52-55.

¹⁰ *Athey v. Leonati*, [1996] 3 S.C.R. 458 458 (“*Athey*”) at para 14; and *Clements v. Clements* 2012 SCC 32 (“*Clements*”) at para. 8.

¹¹ *Clements*, *supra*, at para. 43.

- (a) where the plaintiff establishes that the loss would not have occurred “but for” the negligence of two or more tortfeasors, each possibly in fact responsible for the loss; and
- (b) where the plaintiff through no fault of their own, is unable to show that any one of the possible tortfeasors in fact was the necessary or “but for” cause of the injury, because each can point to one another as the possible “but for” cause of the injury, defeating a finding of causation on a balance of probabilities against anyone.¹²

- [70] These are not the circumstances of this case as there is only one potential tortfeasor – Dr. Martin. As such, causation in this case must be established according to the “but for” test.
- [71] The plaintiffs submit that the “but for” test must be applied in a robust, common sense fashion.¹³ Such that, there is no need for scientific evidence of the precise contribution the defendant’s negligence made to the injury. Evidence connecting the breach of the duty to the injury or loss suffered may permit the judge, depending on the circumstances, to infer that the defendant’s negligence probably caused the loss.¹⁴
- [72] In *Aristorenas*¹⁵, the Court of Appeal explained the proper application of the robust and pragmatic approach set out by the Supreme Court in *Snell* as follows:

Snell and other cases have recognized that it is important to require that the plaintiff provide some sort of evidence (or other considerations) that indicates that the defendant was the cause of the harm suffered. In *Snell*, Sopinka J. at p. 336 S.C.R., 299 D.L.R., disapproved of permitting the plaintiff to “simply prove that the defendant created a risk that the injury which [page 297] occurred would occur”. In *Fairchild v. Glenhaven Funeral Services Ltd.*, [2002] 3W.L.R. 89, [2002] 3 All E.R. 305 (H.L.), Lord Roger of Earlsferry provided this caution about applying the “robust and pragmatic” approach at [para. 150]:

[E]ven though it is always for the judge rather than for the expert witness to determine matters of fact, the judge must do so on the basis of the evidence, including the expert evidence. The mere application of “common sense” cannot conjure up a proper basis for inferring that an injury must have been caused in one way rather than another.¹⁶

¹² *Clements, supra*, at para. 46.

¹³ *Snell v. Farrell*, [1990] 2 S.C.R. 311 (“*Snell*”).

¹⁴ *Snell*, *ibid*, at paras. 32 and 33; and *Clements, supra*, at para.9.

¹⁵ *Aristorenas v. Comcare Health Services*, [2006] O.J. No. 4039 (C.A.) at paras. 56-57, leave to appeal to SCC refused, [2006] S.C.C.A. No. 487 (“*Aristorenas*”).

¹⁶ *Aristorenas, ibid.*, at paras. 63-64.

- [73] A defendant can defeat an inference of causation by advancing some evidence that the plaintiff's loss would have occurred notwithstanding the defendant's breach of the standard of care. As stated by the Supreme Court of Canada in *Clements*:

Where "but for" causation is established by inference only, it is open to the defendant to argue or call evidence that the accident would have happened without the defendant's negligence, i.e. that the negligence was not a necessary cause of the injury, which was, in any event, inevitable. As Sopinka J. put it in *Snell*, at p. 330:

The legal or ultimate burden remains with the plaintiff, but in the absence of evidence to the contrary adduced by the defendant, an inference of causation may be drawn although positive or scientific proof of causation has not been adduced. If some evidence to the contrary is adduced by the defendant, the trial judge is entitled to take account of Lord Mansfield's famous precept [that "all evidence is to be weighed according to the proof which it was in the power of one side to have produced, and in the power of the other to have contradicted" (*Blatch v. Archer* (1774), 1 Cowp. 63, 98 E.R. 969, at p. 970)]. This is, I believe, what Lord Bridge had in mind in *Wilsher* when he referred to a "robust and pragmatic approach to the facts" (p. 569).¹⁷

- [74] Loss of chance is not compensable in medical malpractice cases. Proof that "but for" the defendant's actions the plaintiff had a better chance of avoiding the outcome does not establish causation. The plaintiff must prove on a balance of probabilities, that the outcome would have been avoided.¹⁸

- [75] The Ontario Court of Appeal in *Cottrelle* explained this principle as follows:

...[I]n an action for delayed medical diagnosis and treatment, a plaintiff must prove on a balance of probabilities that the delay caused or contributed to the unfavourable outcome. In other words, if, on a balance of probabilities, the plaintiff fails to prove that the unfavourable outcome would have been avoided with prompt diagnosis and treatment, then the plaintiff's claim must fail. It is not sufficient to prove that adequate diagnosis and treatment would have afforded a chance of avoiding the unfavourable outcome unless that chance surpassed the threshold of more likely than not.¹⁹

ANALYSIS

A. Standard of Care

¹⁷ *Clements*, *supra*, at para. 11 [emphasis in the original].

¹⁸ *Cottrelle v. Gerrard* (2003), 67 O.R. (3d) 737 (C.A.) ("*Cottrelle*").

¹⁹ *Cottrelle*, *ibid.* at para. 25; and see also *Salter v. Hirst*, [2017] O.J. No. 4207 (C.A.) at para. 14, leave to Appeal to S.C.C. refused [2011] S.C.C.A. No. 503.

(i) Expert Evidence

(a) Dr. Brankston

- [76] Dr. Edwin Brankston was qualified as an expert in family medicine. He provided opinion evidence on Dr. Martin's standard of care at the February 10, 2011 health review of Mr. DeMarco. Dr. Brankston was of the opinion that Dr. Martin fell below the standard of care in the medical management of Mr. DeMarco because he failed:
- (i) to conduct and document an adequate systems review;
 - (ii) to document Mr. DeMarco's smoking history;
 - (iii) to assess Mr. DeMarco's cardiac risks other than blood lipids; and
 - (iv) to order a routine ECG.
- [77] Dr. Brankston explained that a periodic health examination involves conducting a review of any changes to the patient's family history, identifying any new allergic or adverse reactions to medications and any current medications that the patient is on. This general inquiry is to be followed by a detailed review of systems, followed by a physical examination and, last, with the development of a plan of management to address any issues identified during the examination requiring investigation or treatment.
- [78] He explained that a "review of systems" (ROS) refers to a review of all of the major organ systems and functions in the body. A "positive" refers to a symptom of concern that requires further clarification leading to a more focused and detailed functional inquiry with respect to the particular symptom.
- [79] Dr. Brankston was critical of Dr. Martin's record-keeping. He noted that Dr. Martin provided no detail with respect to questions he asked of Mr. DeMarco when conducting his review of systems. Without detail, he says he was unable to determine the scope of Dr. Martin's review of systems. Dr. Brankston was of the opinion that a record of a negative review of systems is not helpful in informing the reader of any details with respect to any specific body system.
- [80] Dr. Brankston opined that a general first question would have elicited from Mr. DeMarco the triad of symptoms of which he complained four months later. Dr. Brankston testified that it was the extent of the review of the body's systems by Dr. Martin which was problematic. Dr. Brankston is of the view that Dr. Martin was required to ask specific questions regarding each of the body's main systems – ear, nose, and throat system, respiratory system, cardiovascular system, gastrointestinal system, genitourinary system, neurological system and the musculoskeletal system and joints. With respect to Mr. DeMarco and his history of syncope, Dr. Brankston believes that a detailed inquiry with respect to the neurological system would have elicited from Mr. DeMarco disclosure of his ER visit in July 2010 and his continued feelings of light-headedness and presyncope.

- [81] Dr. Brankston was asked to assume that Mr. DeMarco had been having symptoms of presyncope, dyspepsia, and left arm pain all associated together since July 2010 and provide his opinion with respect to whether Dr. Martin's health review on February 10, 2011 met the standard of care. Based on that assumption, Dr. Brankston was of the opinion that had Dr. Martin asked an appropriate set of questions on the review of systems, the triad of symptoms would have been disclosed by Mr. DeMarco and, furthermore, it was likely that Dr. Martin would have documented those symptoms.
- [82] Dr. Brankston's primary criticism of Dr. Martin was the thoroughness of his functional inquiry. In his opinion, it was highly likely that the triad of symptoms would have come to light at the February 10, 2011 health review had Dr. Martin's functional inquiry been appropriately thorough.
- [83] Dr. Brankston testified that: (i) Dr. Martin indicating with the letter "N" that each of the systems was "negative" (or normal) was insufficient detail of the results of the examination of each body system to satisfy the standard of care of record-keeping on a physical examination; (ii) that Dr. Martin's noting that the review of systems was negative did not meet the standard of record-keeping of the College of Physicians and Surgeons. However, he conceded no evidence of the policies maintained by the College of Physicians and Surgeons was before the court; (iii) it was likely Mr. DeMarco would have brought ongoing symptoms to the attention of Dr. Martin had Dr. Martin obtained a detailed history of his ongoing health symptoms. He believes that, at a minimum, even with normal blood pressure a general physical exam would have revealed the triad of symptoms experienced by Mr. DeMarco; (iv) once the triad of symptoms was disclosed, Dr. Martin should have ordered an ECG; (v) the standard of care mandated that Dr. Martin order a Holter monitor test as well as a referral to a cardiologist for exercise stress testing.
- [84] During cross-examination, Dr. Brankston acknowledged that it is possible for patients suffering from ischemia not to suffer any symptoms. He described this as "silent" ischemia which affects 20% to 30% of all patients with ischemia.
- [85] It was further acknowledged by Dr. Brankston that the conclusion reached by Dr. Martin: "ROS (Review of Systems) – Neg (Negative)", must have been preceded by questions put to and answered by Mr. DeMarco.
- [86] Dr. Brankston admitted during cross-examination that Dr. Martin's failure to document Mr. DeMarco's smoking history did not constitute a breach of the standard of care because Dr. Martin had available to him earlier records identifying Mr. DeMarco as a non-smoker. Furthermore, Dr. Brankston admitted that Dr. Martin had all information he needed available to him to calculate Mr. DeMarco's cardiac risk and that information was not limited to blood lipids. Last, Dr. Brankston admitted that the standard of care does not require a physician to do an ECG on a 51-year-old individual as part of a periodic health examination.

(b) Dr. Stanton

- [87] Dr. Tom Stanton was retained by the defendant to provide his expert opinion on whether Dr. Martin met the standard of care expected of a family physician in 2011. Dr. Stanton was of the opinion that Dr. Martin met the standard of care expected of a community doctor with the exception of his failure to send the referral to Dr. Wong. Dr. Stanton reviewed Mr. DeMarco's medical records as well as the expert reports of Dr. Myers and Dr. Brankston.
- [88] Like Dr. Martin, Dr. Stanton testified that most physicians conduct a review of systems by starting at the top of the head and ending at the bottom of the feet, proceeding in a systematic way through each system and asking a number of questions that flesh out any problem with that particular system. The physician begins with a general question and if the patient identifies specific complaints, the physician follows with questions specific to the identified complaint.
- [89] Dr. Stanton noted that Mr. DeMarco had no known risk of cardiac disease. He was not a diabetic and he was a non-smoker. His cholesterol was at an acceptable level. Dr. Stanton assessed Mr. DeMarco as having a low cardiac risk score of 7.9%.
- [90] Dr. Stanton testified that the triad of symptoms reported in June 2011 were non-specific and not classic for cardiac disease present for over a year. Further support that Mr. DeMarco symptoms were not remarkable or indicative of heart disease is found in the ER records from July 2010. Dr. Stanton noted that an ECG conducted on July 24, 2010 did not reveal results consistent with a cardiac-related issue.
- [91] Dr. Stanton disagreed with Dr. Brankston's assertion that Dr. Martin fell below the standard of care by failing to do an adequate review of systems. Dr. Stanton was of the view that Dr. Martin's review of Mr. DeMarco's medical history together with his physical exam, diagnosis and planned investigation were all performed and documented properly and met the standard of care.
- [92] Dr. Stanton acknowledged during cross-examination that templates are useful tools as they provide a checklist and guard against missing a particular system.
- [93] Dr. Stanton further acknowledged that, had Mr. DeMarco reported the July 2010 fainting episode or recurrent episodes of light-headedness during the February 20, 2011 attendance, a reasonable standard of care would have required Dr. Martin to inquire further with more detailed questions regarding Mr. DeMarco's presyncope.

(ii) Analysis – Fact Finding and Breach of Standard of Care

- [94] The plaintiffs submit that Dr. Martin was negligent in his conduct of the periodic health examination conducted on February 20, 2011 and was negligent in failing to send the stress test referral to Dr. Wong on June 13, 2011. Dr. Martin admits his failure to refer Mr. DeMarco to Dr. Wong for a stress test is a breach of the standard of care. Therefore, the focus of the standard of care analysis is on the periodic health review conducted by Dr. Martin on February 20, 2011.

- [95] The plaintiffs allege that Dr. Martin did not meet the standard of care of a family physician in February 2011 because he failed to ask questions which would have elicited the triad of symptoms.
- [96] Working backwards, the theory of the plaintiffs starts with Mr. DeMarco reporting the triad of symptoms in June 2011 and reporting to Dr. Martin that he had been experiencing the triad of symptoms intermittently since the July 2010 fainting episode. The plaintiffs ask the court to find that on February 10, 2011, Mr. DeMarco was in fact, experiencing the triad of symptoms.
- [97] Dr. Martin testified that had Mr. DeMarco reported the triad of symptoms on February 10, 2011 he would have referred Mr. DeMarco for a stress test. This evidence is corroborated by the fact he did refer Mr. DeMarco for a stress test in June 2011 when the triad of symptoms was first reported.
- [98] Dr. Brankston testified that the first question Dr. Martin should have asked was: “Are there any new issues or symptoms of concern that you’ve experienced since your last visit?” or something similar and if he had asked such a question Dr. Martin would have met the standard of care. Dr. Brankston acknowledged that Dr. Martin may have asked a similar question and Mr. DeMarco simply did not volunteer the information. He recognized that sometimes patients do not report particular symptoms even when prompted.
- [99] Dr. Brankston admitted that he was not asked to and did not consider Dr. Martin’s evidence as to his invariable practice at periodic health examinations in preparing his opinion.
- [100] Furthermore, Dr. Brankston admitted that whatever question was asked of Mr. DeMarco, it elicited the response: “Feeling well except for recurrent back strain”. This response was recorded by Dr. Martin in his notes. Dr. Brankston acknowledged that the usual practice of family physicians is to record only the response of the patient, not the question and response. Dr. Martin testified that his usual practice is to ask as a first question: “How are you feeling?”. I find that was Dr. Martin’s invariable practice and this question, or a similar one, was asked of Mr. DeMarco.
- [101] The evidence establishes that, uncharacteristic of advanced coronary artery disease, Mr. DeMarco did not suffer from any of its classic symptoms. For example, Dr. Martin asked Mr. DeMarco on June 13, 2011 whether he suffered any chest or arm pain on exertion (angina) to which he responded in the negative despite experiencing the triad of symptoms at that time. Aside from his European ancestry, Mr. Demarco was, in every other respect, atypical of an individual suffering from advanced coronary artery disease.
- [102] At the February 10, 2011 physical examination, Dr. Martin did not have knowledge of the fainting episode in July 2010, nor did he have knowledge of Mr. DeMarco’s advanced coronary artery disease. He was not aware that Mr. DeMarco was suffering intermittently from the triad of symptoms. Dr. Martin did, however, have the benefit of Mr. DeMarco’s medical history for the 3 ½ years prior and had concluded Mr. DeMarco

was at low risk of developing coronary artery disease. He was also aware that Mr. DeMarco had a history of vasovagal syncope which was consistent with his fainting episode in July 2010 and that Mr. DeMarco's syncope was not cardiac-related. All of the experts agreed there was nothing significant from a cardiac perspective that occurred between December 2007 and February 2011.

- [103] On February 10, 2011, Dr. Martin had Mr. DeMarco's complete medical history available to him. He had Mr. DeMarco's answers to questions. He had the results of blood pressure, weight, height and heart rate testing. From this information he concluded that Mr. DeMarco was generally healthy. By all outward and objective appearances Mr. DeMarco was healthy.
- [104] Courts must be careful not to rely on the perfect vision afforded by hindsight in order to evaluate a particular exercise of judgment fairly.²⁰ It is within this context and without the benefits that hindsight provides, that Dr. Martin's review of systems on February 20, 2011 must be considered. The plaintiffs submit that the purpose of an annual physical examination is for an otherwise healthy patient to obtain the expert advice of their family physician confirming that, in fact, they *are* healthy. The plaintiffs submit, therefore, that something more than a general initial question about a patient's health is needed to meet the standard of care.
- [105] The plaintiffs acknowledge that in the absence of a specific record or recollection of having conducted the annual physical examination, the court may accept the physician's habitual and invariable practices as evidence of what occurred with respect to a specific physical examination. However, the plaintiffs submit Dr. Martin has a variable rather than a habitual practice for conducting periodic health reviews of his patients. They point to Dr. Martin's evidence that he typically conducts a more fulsome investigation of the cardiac system if a patient is at risk of cardiac disease than if the patient is assessed as low-risk. The plaintiffs argue that this variable approach has the effect of assessing a person's overall health and, specifically, their cardiac health *before* the examination takes place. That is, by reaching a conclusion as to a patient's cardiac health *before* conducting an assessment. In doing so, Dr. Martin breached the standard of care.
- [106] It was only the level and depth of questioning of a patient within the context of Dr. Martin's habitual and invariable practice for conducting periodic health examinations that were subject to modification. Dr. Martin testified that only the degree of specificity in questioning with respect to any particular body system was modified based on his assessment of a patient's overall health. He explained that a patient's health was determined from a review of their medical history and the patient's response to the initial general question of how the patient was feeling.
- [107] I find that Dr. Martin's practice of conducting a review of systems was not variable. He had a specific and consistent approach to his review of systems. He followed the same approach in December 2007 as he did in February 2011. It is noted that Dr. Brankston did not take issue with Dr. Martin's record-keeping with respect to the December 2007

²⁰ *Lapointe v. Hopital Le Gardeur*, [1992] 1 S.C.R. 351 at para. 28.

health examination where he similarly recorded that generally Mr. DeMarco “felt well” and his review of systems was “negative”.

- [108] Mr. DeMarco became a patient of Dr. Martin in January 2007. Dr. Martin was provided with Mr. DeMarco’s medical records from his prior family physician, Dr. Gambrill. His medical history included long-standing vasovagal syncope associated with having blood drawn. His medical history revealed no risk factors for heart disease. Mr. DeMarco’s medical history was, in all respects, unremarkable. The most notable entries consist of his diagnosis and treatment of carpal tunnel syndrome and his diagnosis and treatment of an ankle injury as a teenager.
- [109] The plaintiffs argue that the extent and specificity of the questions asked of Mr. DeMarco were not adequate and fell below the standard of care of a physician practising family medicine in London, Ontario in 2011.
- [110] The review of systems is the initial step in a family physician’s periodic health review. It is followed by a physical examination. Dr. Martin’s usual practice involved asking further questions of his patients during the physical examination as he moved from “head to toe”. Dr. Brankston was not critical of this aspect of Dr. Martin’s health examination of Mr. DeMarco.
- [111] I do not accept the suggestion that Dr. Martin reported conclusions without first asking questions of and receiving responses from Mr. DeMarco. Dr. Martin testified that he would have asked questions of Mr. DeMarco and to the extent any response required further investigation he would have enquired further. I accept Dr. Martin’s evidence of his usual practice in this regard.
- [112] As part of his usual practice, Dr. Martin interviews his patient and determines what next to investigate based upon what the patient reports. Dr. Martin’s usual practice was followed throughout the period of time Mr. DeMarco was his patient. This is evidenced by the medical records of Mr. DeMarco. It is clear from a review of these records that based on the answers provided by Mr. DeMarco, Dr. Martin provided advice, referred him to specialists, ordered testing and prescribed medication, as needed, to address Mr. DeMarco’s symptoms.
- [113] The effectiveness of the practice of every family physician depends upon the patient being forthright with answers to questions. There is no evidence to suggest Mr. DeMarco was not forthright in the answers he gave to the questions asked by Dr. Martin.
- [114] The standard of care of a family physician conducting a review of systems requires only that the physician ask the question of their patient; it does not require that the physician elicit a specific response.
- [115] Here, there is no evidence that Mr. DeMarco was not candid in his responses. Quite the opposite. The evidence clearly establishes that Mr. DeMarco was proactive in seeking medical advice and treatment with respect to his health concerns. Mr. DeMarco was not an individual who downplayed his symptoms or failed to disclose health concerns to his

treating physicians. He sought medical attention in the emergency department following a fainting episode on July 24, 2010. He then sought Dr. Martin's medical advice a few months later in December 2010. He then attended a periodic health examination on February 20, 2011 which was scheduled by Lisa on his behalf. He attended an appointment in May 2011 to address a painful wart. He then scheduled and attended an appointment in June 2011 (without the knowledge of either Lisa or Gabriel) to report the triad of symptoms.

- [116] Although he reported in June 2011 that he had been experiencing the triad of symptoms intermittently since July 2010, the first time Mr. DeMarco reported these symptoms was on June 13, 2011.
- [117] With respect to the fainting episode in July 2010, Mr. DeMarco was discharged after receiving the assurance of a cardiologist that his syncope was not cardiac-related. Mr. DeMarco was advised to return to the hospital if his symptoms recurred. He did not return to the hospital before the examination in February 2011 which suggests his symptoms were not significant. I find, as the medical record states, that on February 10, 2011 Mr. DeMarco reported to Dr. Martin he "generally [felt] well, except recurrent back strain". I find that Mr. DeMarco did, in fact, "feel well" and that he was not suffering from any significant symptoms such that no level of questioning would have elicited the triad of symptoms. This evidence is corroborated by Lisa and Gabriel both of whom confirmed they did not observe the triad of symptoms either before or after the June 13, 2011 attendance.
- [118] Dr. Brankston in providing his standard of care opinion was asked to assume that Mr. DeMarco was in fact experiencing the triad of symptoms on February 10, 2011, however, the evidence does not support this assumption.
- [119] Having found that Mr. DeMarco was not experiencing the triad of symptoms on February 20, 2011 to the same degree as he was on June 13, 2011, the assumption underlying Dr. Brankston's opinion has not been established. Therefore, I do not accept his evidence that Dr. Martin in conducting a review of systems on February 20, 2011 fell below the standard of care.
- [120] The plaintiffs point to the mistakes made by Dr. Martin on June 13, 2011 in failing to send the referral and failing to advise Dr. Wong of Mr. DeMarco's left arm discomfort and ask that I infer Dr. Martin made mistakes when he conducted a review of systems on February 10, 2011 thereby falling below the standard of care. Such an inference is unsupported by the evidence. The mistakes made by Dr. Martin on June 13, 2011 (failing to send the referral to Dr. Wong and failing to mention left-arm discomfort in the referral note) were mistakes unrelated to his review of systems and record-keeping on that same date.
- [121] I find that Dr. Martin's usual practice in conducting periodic health reviews, including his record keeping and his approach to conducting a review of systems, to have been reasonable. Therefore, I find that Dr. Martin's record keeping and review of systems did

not fall below the standard of care of a family physician conducting a periodic health examination in February 2011.

B. Causation

Wait-Time Defence

- [122] As noted earlier in these reasons, the court received the evidence of Lisa with regards to her own experience of wait times for appointments with Dr. Wong following referrals by her family physician. The court also received the evidence of Dr. Wong as to his average wait time for a non-urgent stress test in June 2011. Last, the court received the evidence of Dr. Jablonsky as to his average wait time for an angiogram in summer of 2011. No experts were permitted to testify on the reasonableness of the wait times to which Drs. Wong and Jablonsky testified. Expert evidence is not required to assist the court in making its factual finding as to the average wait time for a referral to Dr. Wong or a referral to Dr. Jablonsky in the Summer 2011.
- [123] Dr. Wong testified that availability of cardiology services varies from year to year based on the number of available cardiologists in London which fluctuates due to retirements and hirings. I find that the average wait time for Lisa's referrals to Dr. Wong in 2008 and 2013 should be given no weight.
- [124] Lisa's evidence with respect to her wait times to see Dr. Wong from November 2011 indicates that Lisa was asked to be placed on a cancellation list which ultimately reduced her wait time from 62 days to 6 days. The original appointment date provided to Lisa is consistent with the average wait time of 59 days for non-urgent referrals to Dr. Wong in June 2011. Mr. DeMarco did not follow-up with Dr. Martin after the June 13, 2011 attendance. This evidence suggests it is not likely that had Mr. DeMarco been advised of an appointment date of July 25, 2011 or August 11, 2011 he would have asked to be placed on the cancellation list.
- [125] Turning now to the weight to be given to Dr. Wong's evidence, it was admitted by Dr. Wong during cross-examination that a patient record may have been missed during the electronic extraction process and/or that the information summarized from the extracted patient records may be incorrect. However, this was admitted only as a possibility. The possibility for human error does not establish that in fact it did occur nor is there evidence that if any such error did occur it would have been statistically significant. Furthermore, the test sample of patient charts reviewed by Dr. Wong and the original appointment date given to Lisa with respect to a non-urgent referral to Dr. Wong in November 2011 serve to bolster the chart's reliability. The plaintiffs have the burden of proof. I note there is no reliable (or any) evidence to support an average wait time of less than 59 days for a non-urgent stress test referral made to Dr. Wong in June 2011. Therefore, I accept Dr. Wong's evidence in this regard.
- [126] With respect to the "wait time defence", I make the following findings of fact:

- (a) Mr. DeMarco did not follow up with Dr. Martin's office, he did not complain to his family regarding the triad of symptoms in the months prior to his death and they did not observe any health issues with Mr. DeMarco in the months prior to his death. His conduct establishes that his symptoms were not worsening and he was not concerned with his health in the months following the June 13, 2011 attendance and leading to the events of August 20, 2011;
- (b) Upon receipt of Dr. Martin's referral, Dr. Wong would have directed that Mr. DeMarco be scheduled for a non-urgent stress test;
- (c) The average wait-time for a non-urgent stress test referral to Dr. Wong in June 2011 was 59 days and, therefore, Mr. DeMarco would have undergone a stress test no earlier than August 11, 2011;
- (d) Mr. DeMarco's pre-test probability for heart disease was low given the absence of any cardiac risk factors and his relatively young age;
- (e) Mr. Demarco's stress test would have been abnormal mid-late stage having regard to the absence of symptoms of angina and his history of regular exercise and manual labour;
- (f) Mr. Demarco would have been referred to Dr. Jablonsky for an angiogram given his low pre-test probability and mid-late stage abnormal stress test results on August 11, 2011;
- (g) The angioplasty referral to Dr. Jablonsky, would have been made on a non-urgent basis having regard to Mr. DeMarco's mid-late stage abnormal stress test considered together with his low pre-test probability;
- (h) The next available date for a non-urgent angiogram referred to Dr. Jablonsky on August 11, 2011 was his clinic following his return from holidays on August 24, 2011;
- (i) The angiogram would have revealed Mr. DeMarco suffered from significant two-vessel coronary artery disease; and
- (j) Dr. Jablonsky would have immediately performed an angioplasty on August 24, 2011.

[127] Although the "wait-time" defence was raised by Dr. Martin, it remains the plaintiffs' burden to establish that "but for" Dr. Martin's failure to send the referral to Dr. Wong, Mr. DeMarco would have received treatment of his advanced coronary artery disease through revascularization prior to August 20, 2011.

[128] I find that Mr. DeMarco would not have received treatment of his coronary artery disease prior to August 20, 2011. Therefore, the plaintiffs have not established on a balance of probabilities that, if the referral note had been sent by Dr. Martin to Dr. Wong on June

13, 2011, Mr. DeMarco would have received revascularization of his coronary artery disease before the events of August 20, 2011.

- [129] Consequently, the plaintiffs have failed to establish that “but for” Dr. Martin’s admitted breach of the standard of care, Mr. DeMarco’s coronary artery disease would have been treated before the events of August 20, 2011.

The Treatment Defence

(i) *Expert Evidence*

(a) **Dr. Myers**

1. Qualifications to Provide Expert Evidence

- [130] Dr. Robert Myers, a cardiologist practising in Toronto, was retained by the plaintiffs to provide his expert opinion on certain questions relating to the issue of causation. In particular, Dr. Myers was asked to provide his opinion with respect to the following issues:

- (a) if Mr. DeMarco had a stress test prior to his massive heart attack in August 2011, would the test have identified that Mr. DeMarco was at risk for this heart attack and in particular, the fact that he had severe atherosclerosis of the left anterior and right coronary arteries with stenosis up to 90%? Is it possible that this would have been diagnosed by Dr. Wong?
- (b) If the risk of the heart attack was identified, what treatment would have been recommended to reduce this risk or address the severe atherosclerosis?
- (c) If this treatment had been pursued, is it possible or probable that a massive heart attack that proved fatal in August 2011 could have been avoided?

- [131] Dr. Myers prepared and served four reports dated May 23, 2013 (the “first report”); January 6, 2014 (the “second report”); March 10, 2017 (the “third report”); and September 11, 2017 (the “fourth report”).

- [132] The defendant challenged the admissibility of Dr. Myers’ evidence, in its entirety, alleging Dr. Myers was biased and not independent; rather, acted as an advocate for the plaintiffs’ position in this litigation. In the alternative, the defendant argued that if Dr. Myers’ evidence is admitted it should be given no or little weight.

- [133] As a result of Dr. Myers’ limited availability to attend in London, I received his evidence both with respect to his qualifications to provide the court with expert evidence and with respect to his expert medical opinion evidence on the issues in this action, without first ruling on the issues of his qualification to testify and the scope of his expert testimony. After receiving Dr. Myers’ evidence, I heard oral submissions of counsel and received written submissions and reserved my ruling on these issues. This is my ruling on Dr.

Martin's motion to exclude Dr. Myers' evidence in its entirety or, in the alternative, to restrict the scope of Dr. Myers' opinion evidence to only that evidence necessary to assist the court in determining the issue of causation.

- [134] I will deal first with the threshold issue of the admissibility of Dr. Myers' evidence. Before addressing the evidence and the specific issues raised on this *voir dire*, I will outline the key principles of law.
- [135] The seminal decision on the admissibility of evidence is the Supreme Court decision in *R. v. Mohan*.²¹ Expert evidence will be admitted if it is relevant - this requires a finding that the evidence is logically relevant to the issues in the case and the benefits of the evidence (in terms of materiality, weight and reliability) outweigh its costs (both practical and prejudicial); it is necessary and provides information likely to be outside the experience and knowledge of the trier of fact; the expert is properly qualified; and there is no applicable exclusionary rule.
- [136] The *Mohan* test was refined by the Supreme Court in *White Burgess*.²² At the first stage of the inquiry, there are four threshold preconditions that the proponent of the evidence must establish in order for the proposed expert opinion to be admissible: (1) relevance; (2) necessity in assisting the trier of fact; (3) absence of an exclusionary rule; and (4) a properly qualified expert.²³
- [137] The second stage of the inquiry involves a cost-benefit analysis of the issue of whether otherwise admissible expert evidence should be excluded because its probative value is outweighed by its prejudicial effect. This second stage has been interpreted as the Court's "general residual discretion" to exclude evidence that is more prejudicial than it is probative - also described as the Court's "gatekeeper" function.²⁴
- [138] The defendant argues that Dr. Myers' proposed evidence fails at the first stage of the analysis because the content of his third and fourth reports demonstrate he has assumed the role of an advocate and is therefore neither independent nor impartial. Prerequisites for his proper qualification as an expert.
- [139] The primary duty of any expert witness is to be fair, objective and to avoid being partial to the position of any particular party to the litigation. The expert has an obligation to the court and not to the party who retained him or her.²⁵ In Ontario, the *Rules of Civil Procedure*²⁶ were amended in 2010 to impose distinct duties on all witnesses who

²¹ [1994] 2 S.C.R. 9 ("*Mohan*") at para. 17.

²² *White Burgess Langille Inman v. Abbott and Haliburton Co.* 2015 SCC 23 ("*White Burgess*") at paras.17-21; and *R. v. Mohan*, *supra* note 3 at paras. 52-54.

²³ *Mohan*, *supra* note 3 at paras. 20-25.

²⁴ *Bruff-Murphy v. Gunawardena*, 2017 ONCA 502 at para. 36 ("*Bruff-Murphy*").

²⁵ *White Burgess*, *supra* at para. 49.

²⁶ R.R.O. 1990, Reg. 194.

propose to provide expert opinion evidence.²⁷ One of those duties requires the expert to provide only fair, objective, and non-partisan opinions.

- [140] Rule 53.03 requires the expert to sign an acknowledgement of expert's duty whereby the expert acknowledges that it is his or her duty to provide opinion evidence that is fair, objective and nonpartisan; to provide opinion evidence related only to matters that are within the expert's area of expertise; and to provide such additional assistance as the court may reasonably require to determine a matter in issue. The expert must further acknowledge that the duties imposed upon him or her prevail over any obligation which the expert owes to any party to the litigation.
- [141] Dr. Myers executed an Acknowledgement of Expert's Duty. During his examination-in-chief, he stated that he understood his duties and denied assuming the role of advocate for the plaintiffs.
- [142] In support of the motion to exclude the evidence of Dr. Myers, the defendant made the following arguments:

- i. having been retained by the plaintiffs to provide an opinion on questions relating to the issue of causation, Dr. Myers did not respond to the specific question asked of him rather provided the following opinion in his first report:

In my opinion, the standard of care was breached in that 9 ½ weeks elapsed between Mr. DeMarco's original visit with Dr. Martin, at which time he suspected coronary artery disease, and Mr. DeMarco's death. Had Mr. DeMarco been seen by a cardiologist and/or had an exercise stress test prior to his death, the severe coronary artery disease identified on autopsy would likely have been diagnosed and treated and Dr. DeMarco would not have died suddenly.

The defendant submits that Dr. Myers, a cardiologist, was not retained to provide an opinion on the standard of care of Dr. Martin, a family physician, and does not possess the necessary qualifications to give opinion evidence on the standard of care of a family physician and therefore his opinion evidence is inadmissible.

- ii. In the third and fourth reports prepared in response to the responding reports of Dr. David Fitchett, a cardiologist, and Dr. Jablonsky, Dr. Myers uses partisan language and levels personal attacks at the defendant's experts. Examples are:

“In my opinion, Dr. Jablonsky's discussion about stress testing and the likely results of a stress test in Mr. DeMarco is nonsensical.”

²⁷ Rule 4.1.01.

“...Remarkably, he speculates that Mr. DeMarco’s ‘right ankle issues as well as back pain’ may have resulted in an inability to complete the test and thus delay diagnosis.”

“Even in Dr. Jablonsky’s speculative narrative of a 6-8 week delay to see Dr. Wong and 1 to 3 week wait to have an angiogram...”

“Inexplicably, Dr. Jablonsky believes that an abnormal stress test results was uncertain.”

“Throughout his report, Dr. Jablonsky detracts from the impact of Mr. DeMarco’s death nearly two and half months after a consultation should have been initiated, by accumulating multiple potential delays, from the stress test limited by ankle and back problems, two possible delays in arranging further non-invasive tests, to his own holiday and attending the hospital schedule.”

“... The intent of Dr. Jablonsky appears to be the use of all potential delays to justify his conclusion that Mr. DeMarco’s death was inevitable. A reasonable physician would realize that more likely than not, Mr. DeMarco would have been reviewed by a cardiologist earlier, had an abnormal stress test and lifesaving cardiac intervention.”

...

[143] The defendant argues that notwithstanding the inflammatory language used, Dr. Myers failed to answer the specific questions he was asked in the letter of instruction making his evidence inadmissible on the basis it is merely dressed-up advocacy and does not assist the court.

[144] I find that Dr. Myers is able to provide assistance to the court and is not biased so as to disqualify him from providing expert evidence in this case. This is not the “clearest of cases”. I accept that he was able to perform his duties as an expert. I find the language he used to express his contrary views and opinions to those of the defendant’s experts was intended simply to underscore the strength with which he holds those views and opinions. The language is strong and unhelpful to the court but does not go so far as to constitute a personal attack on either Dr. Fitchett or Dr. Jablonsky.

[145] Having ruled that Dr. Myers’ evidence is generally admissible. I now turn to the scope of the areas in which Dr. Myers has expertise necessary to qualify him for purposes of providing expert evidence to the court.

[146] The plaintiffs seek to qualify Dr. Myers as an expert in cardiology and specifically in the following areas:

(a) the causes of heart disease;

- (b) the causes of cardiac related death;
- (c) the diagnosis of heart disease; and
- (d) the treatment of heart disease including the treatment of coronary artery disease.

[147] The defendant objects to Dr. Myers' qualification to provide expert evidence as to the mechanism of the fatal arrhythmia suffered by Mr. DeMarco; however, does not object to Dr. Myers' qualification to provide expert evidence on the causes of sudden cardiac-related death, generally. The defendant also objects to Dr. Myers' qualifications to provide expert evidence on the treatment of coronary artery disease. To support his objections, the defendant points out that Dr. Myers is neither an interventional cardiologist nor a cardiac pathologist.

[148] During direct and cross-examination on his qualifications, Dr. Myers testified as follows:

- (a) he does not have the same expertise as an expert in angiography; however, he routinely reviews results of angiogram tests performed on his patients to inform his treatment of patients who might be identified as having coronary artery disease;
- (b) he has never performed an angioplasty;
- (c) he will refer patients to an interventional cardiologist for an angiogram;
- (d) to the extent any of his patients require angiograms or angioplasty or consultation with an interventional cardiologist, he refers them to an interventional cardiologist;
- (e) he does not discuss the risks and potential complications associated with either an angiogram or angioplasty with his patients;
- (f) the interventional cardiologist has the ultimate decision as to whether he or she is going to conduct the procedure; however, he cannot recall in 25 years' of practice an interventional cardiologist not proceeding upon his recommendation that a patient have an angiogram;
- (g) he is not qualified to decide whether his patient requires an angioplasty rather that decision is made between his patient and the interventional cardiologist;
- (h) he would defer to the interventional cardiologist as to how to proceed in the case of identifying coronary artery disease and treatment options; and
- (i) he admitted that cardiac pathology is not within his area of expertise.

[149] In his fourth report Dr. Myers opines that Mr. DeMarco's death was caused by severe coronary artery disease. He believes that had he been diagnosed after his visit with Dr.

Martin in February 2011, six months before his death, he would have undergone a stress test followed by a coronary angiogram and lifesaving revascularization with bypass or angioplasty.

- [150] The fact that Dr. Myers does not himself conduct revascularization procedures, does not disqualify or preclude him from testifying as to what the procedure entails and whether it is an appropriate treatment of coronary artery disease, generally.
- [151] Dr. Myers has extensive experience practising cardiology and continually and regularly reviews literature relating to cardiology, generally, as well as literature in cardiology subspecialty areas. While not performing the procedures themselves, Dr. Myers follows his patients through cardiovascular treatment, and is primarily responsible for the patient's follow-up care. From this experience and education, Dr. Myers is able to provide expert evidence to assist the Court with respect to the effectiveness of the various treatments in treating coronary artery disease.
- [152] The plaintiffs sought to elicit from Dr. Myers his opinion with regards to the average wait time for a stress test following a patient's referral to a cardiologist by their family physician as well as the average wait time for an angiogram following an abnormal stress test. Before testifying at this trial, Dr. Myers had not previously visited the City of London. Suffice to say, Dr. Myers has never practised cardiology in the City of London. This case involves the wait time for a stress test based on a referral to Dr. Wong, a cardiologist practising in London. Dr. Wong is not a defendant in this action. Whether the average wait time for a stress test for patients referred to Dr. Wong in June 2011 fell below the standard of care of a cardiologist practising in the City of London is not an issue before the court. Therefore, evidence of the average wait times for a stress test and/or an angiogram in the City of Toronto is irrelevant.
- [153] The average wait time for a stress test and, thereafter, an angiogram/angioplasty had the referral been made to Dr. Wong on June 13, 2011 are factual findings. No expert evidence is needed to assist the court in making these findings.
- [154] Dr. Myers is not trained in and does not practise family medicine. Dr. Myers is not trained in and does not practise in forensic pathology. I find, therefore, that Dr. Myers is not qualified to provide evidence on the standard of care of a family physician. Dr. Myers is similarly not qualified to interpret the pathology report of Dr. Chakrabarti and provide expert evidence on the mechanism of Mr. Demarco's sudden cardiac death.
- [155] Based on his education, training and experience, I find that Dr. Myers is qualified to provide expert evidence on the treatment of coronary artery disease, the manner in which a stress test is conducted and whether Mr. Demarco would likely have had an abnormal stress test if he had been tested. Last, Dr. Myers is qualified to provide expert evidence on the causes of arrhythmia, generally, however he is not qualified to provide expert evidence on the cause of Mr. DeMarco's fatal arrhythmia, specifically.

2. Admissible Evidence of Dr. Myers

- [156] I now turn to Dr. Myers' evidence in the areas in which he has been qualified. The early portion of his testimony focussed on medical terminology to which all causation experts referred in their respective testimony. Medical terms such as "ischemia" and "angina" were explained. In addition, Dr. Myers assisted the court in describing the various components of the heart and their function using lay terminology and helpful analogies. Of particular assistance to the court was the introduction through Dr. Myers of a diagram²⁸ describing the four possible causes of Mr. DeMarco's fatal arrhythmia (sudden cardiac death) also referred to as the four possible "mechanisms of death". These four "options" formed the foundation for the opinions of the 5 causation experts – Drs. Myers, Chaisson, Butany, Jablonsky and Fitchett.
- [157] It was agreed by the causation experts that the four possible mechanisms/causes of Mr. DeMarco's sudden cardiac death were:
- Option A: unstable plaque in the narrowed arteries (the wall of the artery is injured) leading to electrical instability in the heart muscle. This can occur at rest.
 - Option B: a blood clot causing a complete blockage of blood flow and leading to a myocardial infarction (heart attack) or thrombosis causing electrical disruption.
 - Option C: scarring (caused by prior ischemic events) leading to electrical instability in the heart muscle.
 - Option D: ischemic cardiomyopathy - the heart is weakened from reduced blood flow to the heart over a period of years leading to a deterioration of the electrical system.
- [158] In his fourth (and final) report, Dr. Myers concluded that in his opinion Mr. DeMarco died as a result of severe coronary artery disease. He further concluded that had Mr. DeMarco been diagnosed in February 2011 he would have undergone a stress test followed by coronary angiogram and lifesaving revascularization with bypass or angioplasty. Dr. Myers also says he would have been prescribed lifesaving medications.
- [159] Dr. Myers testified that beta blockers are effective treatment for arrhythmias because they act to reduce the oxygen demands of the heart by lowering heart rate and blood pressure.
- [160] Dr. Myers acknowledged that Mr. DeMarco had no evidence of a blood clot in his coronary arteries and although Mr. DeMarco had narrowing of his arteries, his arteries remained open. He admitted Mr. DeMarco's heart muscle function was normal with no evidence of multiple heart attacks. He further agreed there were small microscopic foci of scar tissue found during the autopsy.

²⁸ Exhibit "H".

- [161] Dr. Myers testified that it was more likely than not Mr. DeMarco would have had significant findings following a stress test. And, further, that it was more likely than not Mr. DeMarco would have undergone an angiogram following his abnormal stress test and would likely have gone on to have some form of intervention such as an angioplasty.
- [162] Dr. Myers admitted it was difficult to say what exactly happened to Mr. DeMarco but he believed it was likely a combination of factors. However, he could not identify the most significant factor. He speculated whether it was an ischemic element from Mr. DeMarco's very narrowed arteries, or whether a small platelet formed that was undetectable on autopsy, or whether the microscopic scarring caused the arrhythmia or whether a combination of factors interacted together.
- [163] Dr. Myers struggled to say which was the most likely cause of Mr. DeMarco's sudden death among the possible individual factors or combination of factors. However, by the end of his testimony Dr. Myers opined that, the mechanism of Mr. DeMarco's fatal arrhythmia was transient ischemia (despite the absence of any unstable plaque). He further opined that if Mr. DeMarco had a diagnosis of coronary artery disease and had been revascularized by angioplasty, it was more likely than not Mr. DeMarco would not have had a sudden death cause by transient ischemia on August 21, 2011.
- [164] Having ruled that Dr. Myers is not qualified to give evidence on the mechanism of Mr. DeMarco's death, I have disregarded his evidence with respect to the mechanism of Mr. DeMarco's fatal arrhythmia on August 20, 2011 and whether treatment of transient ischemia would have prevented his death.
- [165] It was admitted by Dr. Wong during cross-examination that a patient record may have been missed during the electronic extraction process and/or that the information summarized from the extracted patient records may be incorrect. However, this was admitted only as a possibility. The possibility for human error does not establish that in fact it did occur nor is there evidence that if any such error did occur it would have been statistically significant. Furthermore, the test sample of patient charts reviewed by Dr. Wong and the original appointment date given to Lisa with respect to a non-urgent referral to Dr. Wong in November 2011 serve to bolster the chart's reliability. The plaintiffs have the burden of proof. I note there is no reliable (or any) evidence to support an average wait time of less than 59 days for a non-urgent stress test referral made to Dr. Wong in June 2011.

(b) Dr. Butany

- [166] Dr. Jagdish Butany is a cardiac pathologist. He was qualified as an expert in cardiac pathology for the plaintiffs. He was qualified to provide his expert opinion on:
- (a) the causes of heart disease;
 - (b) the causes of cardiac related death;
 - (c) the effect of cardiac treatment on the heart including the coronary arteries;

(d) the mechanism of Mr. DeMarco's sudden death; and

(e) whether treatment of Mr. DeMarco's coronary artery disease would have prolonged his life.

- [167] Dr. Butany appropriately identified in his Acknowledgement of the Expert Duty that he did not have specific training and education in the area of interventional cardiology. He admitted to the limitations in his opinion with respect to the issue of whether treatment of Mr. DeMarco's coronary artery disease would have prolonged his life. He candidly acknowledged that to the extent his opinion differed on this issue from the opinion of an interventional cardiologist, the court should defer to the latter.
- [168] Dr. Butany explained that the first step in an autopsy is a gross examination which involves holding the heart in your hand and observing it with your naked eye. The pathologist measures the chambers of the heart, the thickness of the heart wall, the valve sizes and whether abnormalities exist in the arteries. Sections are then taken of the heart. The heart is weighed and any externally evident abnormalities are identified.
- [169] Dr. Butany described the condition of coronary artery disease. In young people, coronary arteries are smooth tubes made of muscle. The surface of the wall is the "lumen". As an individual ages, the smooth surface of the wall becomes bumpy and thicker. Fat deposits form and later these irregular areas develop plaque. As plaque continues to grow over time it narrows (stenoses) the vessel. Plaque is also known as atherosclerosis. Complicated plaque in the coronary arteries leads to heart damage.
- [170] A significant build-up of plaque causes narrowing of the inside the artery thereby reducing blood flow regardless of the demands of the heart. Upon exertion, a person with significant narrowing or stenosis of the arteries will develop symptoms of ischemia (lack of blood/oxygen) such as pain in the chest. If the ischemia is sustained, the person will suffer a heart attack otherwise known as a "myocardial infarction" resulting in damage (fibrosis or scarring) to the heart muscle
- [171] As people age and suffer ischemic events, damage (scarring) of the heart muscle occurs which can lead to an arrhythmia. An arrhythmia is a disturbance of the normal rhythm of the heart muscle. All experts agreed that, in addition to heart attacks, ischemia can lead to arrhythmias.
- [172] Dr. Butany explained that, at rest, the heart typically beats on average of 72 beats per minute which allows the heart time to relax, fill with blood and then eject that blood. During ventricular fibrillation, the heart muscle is fibrillating (quivering) and contracting at 300-400 times per minute preventing blood from filling the chambers of the heart leaving no blood to eject. Death occurs quickly during ventricular fibrillation.
- [173] Myocardial scarring is caused by the death of healthy heart muscle/tissue. Scarring can range from tiny microscopic scarring to transmural (or replacement) scarring visible on gross examination. A transmural scar is related to a complete occlusion of a coronary artery such as a myocardial infarction (heart attack).

- [174] Dr. Butany spoke about coronary vasospasms. A coronary vasospasm occurs within the arteries feeding the heart and can lead to a sudden narrowing of the lumen of the vessel thereby reducing or eliminating the blood supply to the heart muscle. Spasms can occur in the presence of atherosclerotic plaque which causes further narrowing. A sudden spasm in an already narrowed artery can completely occlude the vessel leading to ischemic damage to areas of the heart muscle served by that artery.
- [175] A spasm can also cause plaque to tear forming small thrombi which launch downstream resulting in small scars. Coronary vasospasm is a transient condition that can come and go without detection. It usually occurs when a person is upset or experiences a sudden fear or sudden excitement. Dr. Butany explained that plaque disruption caused by spasms occurs at points in the artery which are less diseased than those which are severely diseased or significantly diseased. In arteries having between 40 to 60% narrowing and having hot plaque or unstable plaque, the plaque in these areas becomes thinner because the body cells just under the plaque are gradually eating the plaque from underneath. Thin plaque is vulnerable and susceptible to tear or erosion. A sudden surge in blood caused by excitement or fear can cause thinning plaque to rupture, leading to complete occlusion of the artery resulting in a myocardial infarct.
- [176] After this general overview, Dr. Butany turned his evidence to the autopsy findings of Mr. DeMarco.
- [177] Dr. Butany suggested that the paler areas of Mr. DeMarco's heart observed on gross examination could be areas of ischemic damage (recent damage not scar tissue) or artifactual – nothing of significance.
- [178] Dr. Butany believed that Mr. DeMarco's coronary artery disease was slightly less than 90% and closer to 80% although stated that anything above 75% constitutes severe atherosclerosis. He said there was no evidence of an acute myocardial infarct; however, in the case of a sudden acute myocardial infarct, myocardial ischemic damage is virtually impossible to identify at autopsy because the morphologic changes associated with the myocardial infarction take time to develop. At 12 to 18 hours and certainly at 24 hours and beyond, there would be no difficulty picking up the area of infarction both on gross examination and on the histologic examination.
- [179] Dr. Butany believed that the contraction band necrosis was related to resuscitative efforts rather than an ischemic event. The old healed microscopic scars on the ventricles were likely caused by previous episodes of vasospasm, or small transient ischemic events or small thrombo emboli.
- [180] Dr. Butany acknowledged that small foci where the normal muscle fibers have been lost and replaced with the scar tissue can result in an abnormal rhythm such as ventricular fibrillation.
- [181] Dr. Butany reviewed the nine slides of Mr. DeMarco's heart. He was of the opinion that the extent of scarring in Mr. DeMarco's heart was very mild. There was no evidence of transmural scarring which he explained involves at least half the thickness of the wall of

the myocardium. He noted that in Mr. DeMarco's heart, much of the scarring was interstitial between muscle fibers – meaning the scarring was interspersed between healthy muscle tissue. This scarring would have existed prior to the events of August 20, 2011. Dr. Butany testified that the level of scarring evident from the histological slides may have appeared as paler areas on gross examination.

[182] Dr. Butany confirmed the following:

- (a) there was evidence of severe coronary artery disease;
- (b) there was no evidence Mr. DeMarco suffered an acute coronary event i.e., a heart attack;
- (c) there was no evidence of thrombosis, hot plaque or inflamed plaque or a disruption of plaque or significant inflammation;
- (d) the ambulance and ER records confirm Mr. DeMarco's heart was beating at a rate consistent with ventricular fibrillation; and
- (e) Mr. Demarco died as a result of suffering ventricular fibrillation (fatal arrhythmia).

[183] In cross-examination Dr. Butany admitted/acknowledged:

- (a) It takes about 6 to 8 weeks for a scar to mature. Mr. DeMarco's scars were historic scars or old scars which tells us the ischemic events leading to the scars likely occurred prior to his June 13, 2011 attendance;
- (b) You cannot treat scar tissue;
- (c) Based on the autopsy results, the dyspepsia that Mr. DeMarco was suffering from in June 2011 was not cardiac in origin;
- (d) There was no evidence of thrombosis in the slides of the coronary arteries reviewed;
- (e) There was no evidence of hot plaque or unstable plaque and, in fact, the plaque was fairly stable;
- (f) There was no evidence of an occlusion;
- (g) There was no evidence of an ischemic cardiomyopathy;
- (h) Scarring in and of itself can cause an arrhythmia - even a microscopic scar can cause rhythm problems;
- (i) Without evidence of unstable plaque and thrombosis, Option A is unlikely;

(j) Ventricular hypertrophy can cause an arrhythmia in and of itself. Mr. DeMarco had evidence of this in his left ventricle. Stenting and angioplasty or bypass surgery would not correct this problem;

(k) In any individual case, the literature counts for almost nothing; and

(l) A vasospasm cannot be prevented. There is no preventative treatment.

[184] Dr. Butany postulated that Mr. DeMarco's sudden cardiac death was caused by a vasovagal spasm or that the plaque ruptured but was not observed on autopsy because the segment of the artery where the rupture occurred was not sectioned. He also explained it was possible that thrombus may have been disrupted and dislodged. He described these as possibilities and not likelihoods.

[185] Dr. Butany fairly acknowledged that:

(a) he could only speculate that there was thrombosis in the areas of Mr. DeMarco's coronary arteries that were not sectioned and that he could not say with certainty there was or was not any thrombosis in the rest of the coronary artery tree; and

(b) it was possible that Mr. DeMarco had suffered a vasospasm - the muscle around Mr. DeMarco's coronary arteries was intact and therefore able to spasm.

[186] Dr. Butany's final opinion as to the most likely scenario causing Carlo DeMarco's sudden death was: "Given that I don't have any evidence for a thrombosis I would go to vasospasm."

[187] Dr. Butany testified that in his view small foci of scarring were not enough on their own to cause a fatal arrhythmia but agreed that there was no evidence of other ischemic events to combine with microscopic scarring to cause a fatal arrhythmia. Therefore, he concluded that the mechanism of death was a vasospasm because it would have left no evidence behind.

[188] It was Dr. Butany's overall opinion that, based on the microscopic scars (scar tissue or fibrosis interspersed with healthy muscle tissue) and Mr. DeMarco's severe atherosclerosis, the likely mechanism of Mr. DeMarco's death was that he died of coronary artery disease with occlusion and damage to the muscle (the damage was not evident on autopsy because the interval between death and autopsy may have been too short.) In his opinion, Mr. DeMarco had either an occlusion of an artery not evident on autopsy or a vasospasm superimposed on his significant disease that led to ischemia which resulted in a fatal arrhythmia.

(c) **Dr. Chaisson**

- [189] Dr. David Chaisson, a forensic pathologist, was retained by the defendant to provide an opinion on causation and, in particular, an opinion as to the likely cause and mechanism of Mr. DeMarco's sudden death. Dr. Chaisson was qualified as an expert in the field of forensic pathology and cardiovascular pathology with expertise in sudden unexpected death including sudden cardiac death.
- [190] Dr. Chaisson, testified as to the adequacy of the autopsy performed by Dr. Chakrabarti. He noted that the heart was sampled more extensively than he would have in the circumstances. He was of the view that Dr. Chakrabarti did a very good job of sampling the coronary arteries - he looked at the three major arteries – the circumflex, the left anterior descending and the right coronary. Upon noticing fibrosis on the initial sections sampled, Dr. Chakrabarti went back and took more heart tissue samples.
- [191] Dr. Chaisson testified that had there been evidence of a thrombus, a clot within a coronary artery, or evidence that there might have been some plaque hemorrhage, or when artery cutting through you would sample those areas as well. Dr. Chaisson was of the view these would be grossly visible.
- [192] Dr. Chaisson noted that all of the plaque in the arteries appeared to be stable and found no evidence of acute ischemic injury or any other acute pathology.
- [193] With respect to the extent of the scarring, Dr. Chaisson noted the following:
- (a) patchy fibrosis (extensive myocardial scarring) extending from the inner zone to the outer zone;
 - (b) the patchy fibrosis was transmural because it involved the full thickness of the wall;
 - (c) the damage was through the thickness of the ventricular wall and extended beyond the subendocardial area;
 - (d) the patchy fibrosis indicated significant ischemic injury;
 - (e) the more fibrosis the greater risk of rhythm disturbance;
 - (f) the scarring would have been observable on gross examination; and
 - (g) the existence of multiple patchy, old healed scars in the left ventricle.
- [194] On cross-examination, Dr. Chaisson clarified that his reference to transmural scarring was not a reference to replacement scarring in the sense that all of the muscle has essentially died; rather, only that the patchy fibrosis was observed through the thickness of the wall. He acknowledged that the scarring on Mr. DeMarco's heart muscle was not replacement-type scarring, rather only interstitial - interspersed between healthy heart muscle.

- [195] Dr. Chaisson explained that when a person has myocardial scarring like Mr. DeMarco, the electrical impulses run into the scar tissue which act as roadblocks for the impulses. He could not explain why suddenly these foci of fibrosis become disjointed and cause an arrhythmia - “but they do”. Like Dr. Butany, Dr. Chaisson agreed that small foci can cause arrhythmias.
- [196] Dr. Chaisson was of the opinion, to a reasonable degree of medical certainty, it was more likely than not that the mechanism of Mr. DeMarco’s death was a cardiac arrhythmia precipitated by myocardial fibrosis (scarring) and that this fibrosis was the result of remote episodes of ischemia of the myocardium. This ischemia was due to Mr. DeMarco’s underlying severe coronary artery atherosclerosis.
- [197] Dr. Chaisson opined that Option C (scarring) led to the cardiac arrest. His opinion was based on the autopsy results showing multiple areas of scarring on Mr. DeMarco’s myocardium. In support of his opinion, he noted there was no evidence of unstable plaque necessary for Option A, there was no evidence of an acute coronary thrombus or occlusion of the coronary arteries required to support Option B, and there was no evidence of an acute (recent) ischemic injury. There was only evidence in support of Option C.
- [198] Dr. Chaisson disagreed with Dr. Butany that Mr. DeMarco’s ventricular fibrillation was caused by a coronary spasm and stated: “I see no reason to think that coronary spasm had any role to play in the death.” Dr. Chaisson explained that a spasm cannot be proven or disproven on autopsy. He further explained that, typically, you turn to a spasm as a way of explaining a sudden cardiac death in circumstances where there is ischemic damage and there is no severe coronary artery disease. Here, there is no question about severity of Mr. DeMarco’s coronary artery disease. Dr. Chaisson further explained in support of his opinion as to the mechanism of death that scarring would have remained and the longer scarring was present there was an ongoing risk of cardiac arrhythmia because scarring is a known cause of ventricular fibrillation.

(d) Dr. Jablonsky

- [199] Dr. George Jablonsky testified in support of the defendant’s case. He is an interventional cardiologist - a cardiologist who does invasive procedures, and has the ability to carry out interventional procedures, specifically coronary angioplasties.
- [200] Dr. Jablonsky was qualified as an expert in interventional cardiology with expertise in the management and treatment of coronary artery disease. He was permitted to give opinion evidence on the effects and management of treatment for coronary artery disease and the likely cause of Mr. DeMarco’s death.
- [201] Dr. Jablonsky noted that treatment by medical management, angioplasty or coronary bypass would not have addressed Mr. DeMarco’s pre-existing scars. Dr. Jablonsky testified that angioplasty only serves to improve blood flow thereby ameliorating ischemia. However Mr. DeMarco’s sudden death was related to the scarring rather than acute ischemia and so it would have been impossible to know in advance that he had

scarring. He testified: “In this case we have retrospective evidence that the patient already had scarring which put him at risk of sudden cardiac death.”

- [202] Dr. Jablonsky clarified during cross-examination that aggressive medical management or angioplasty are equally effective in extending life based on the literature but only in symptomatic patients. He noted that Mr. DeMarco was not symptomatic of coronary artery disease.
- [203] Dr. Jablonsky opined that revascularization only reduces mortality in patients with certain patterns of coronary artery disease. These situations are left main coronary artery disease or triple-vessel coronary artery disease with impaired left ventricular function. Mr. DeMarco did not have left main coronary artery disease. Mr. DeMarco did not have triple vessel coronary artery disease with impaired left ventricular function. Mr. DeMarco had two vessel coronary artery disease with likely normal ventricular function.
- [204] With respect to the issue of causation, Dr. Jablonsky admitted that with respect to the possible mechanisms of death:
- (a) some patients die of sudden cardiac death as a result of significant scar tissue which creates a fatal arrhythmia in the absence of an ischemic event at that time;
 - (b) some patients have an acute occlusion without any scar tissue on their hearts and that can lead to a fatal arrhythmia; and
 - (c) some patients have some fibrosis and an acute ischemic event and those things combine to create a fatal arrhythmia.
- [205] Dr. Jablonsky was of the opinion that the mechanism of Mr. DeMarco’s death was Option C - scarring.

(e) Dr. Fitchett

- [206] Dr. David Fitchett is a cardiologist who was retained by the defendant to provide an opinion on the issue of causation and Mr. DeMarco’s life expectancy. He was qualified to provide expert evidence in cardiology with expertise in cardiac death and the management and treatment of coronary artery disease.
- [207] Dr. Fitchett explained that Mr. DeMarco was asleep or at rest when he developed ventricular fibrillation and therefore his myocardial oxygen demands would have been low. A beta blocker is a drug which interferes with the effect of adrenaline on the heart. A beta blocker will prevent the sudden death associated with exertion. However, Mr. DeMarco’s heart rate was slow when he developed ventricular fibrillation. Dr. Fitchett concluded that beta blockers would have been of no assistance.

- [208] Dr. Fitchett opined that Mr. DeMarco died suddenly in his sleep as a result of developing ventricular fibrillation and, as such, his death would not have been prevented by coronary revascularization (or sending the referral note to Dr. Wong).
- [209] Dr. Fitchett testified that stenting of DeMarco's two arteries that had 80 or 90% stenosis would have had no impact on other arteries that had less stenosis and any potential area of instability in those less stenosed arteries.
- [210] Of the four possible causes of Mr. DeMarco's fatal arrhythmia, Dr. Fitchett concluded that Option C was the most probable cause. In his opinion, none of the medical treatments or revascularization procedures recommended by Dr. Myers would likely have prevented Mr. DeMarco's death or prolonged his life because the arrhythmia was caused by scarring which is not treatable.
- [211] On cross-examination Dr. Fitchett agreed that the greater amount of scarring, the more likely to have an arrhythmia. Conversely, the less scar tissue, the less likely a person would develop an arrhythmia.
- [212] Dr. Fitchett further agreed that Mr. DeMarco would have been referred for an angiogram which would have revealed two-vessel coronary artery disease and coronary stents would have been deployed. He agreed that Mr. DeMarco was having episodes of transient ischemia in the time prior to his death which caused the scarring observed on autopsy.
- [213] Dr. Fitchett acknowledged that there are a significant number of individuals who suffer a fatal arrhythmia caused by an interaction between scarring and an ischemic episode. Dr. Fitchett admitted that his interpretation of the studies upon which he relied in providing his opinion was incorrect and that he had overstated the results of the studies in favour of the defence position.

(ii) *Analysis and Fact Finding*

- [214] The statement of claim prior to its further amendment at the commencement of trial, was prepared based on the mistaken belief that Mr. DeMarco had suffered a massive heart attack which led to his sudden death. It was not until September 2017, that the plaintiffs first indicated their intention to amend their claim to plead that the mechanism of death was not a massive heart attack; rather, was a fatal arrhythmia caused by Mr. DeMarco's untreated advanced coronary artery disease.
- [215] The defendant argues that each of the "wait time defence" and the "treatment defence" provides a complete defence to the plaintiffs' claims. I have accepted the "wait time defence"; however, for completeness, I will also consider and determine the merits of the "treatment defence".
- [216] The central issue is whether the ventricular fibrillation (fatal arrhythmia) suffered by Mr. DeMarco was caused by the failure to treat Mr. DeMarco's coronary artery disease as alleged by the plaintiffs.

[217] To summarize the expert evidence as to the mechanism of Mr. DeMarco's death:

- (a) Dr. Butany was of the opinion that microscopic/interstitial scarring (Option C) on its own was unlikely to be the cause of Mr. DeMarco's sudden cardiac death and that a combination of Options B (vasospasm) and C (scarring) or Option B on its own were possible causes of Mr. DeMarco's sudden cardiac death; and
- (b) Drs. Chaisson, Jablonsky and Fitchett were of the opinion that microscopic/interstitial scarring (Option C), on its own, was the likely cause of Mr. DeMarco's sudden cardiac death;

[218] All experts agreed with Dr. Myers' opinion as to the four causes of fatal arrhythmia. All experts agreed that microscopic scarring on Mr. Demarco's heart revealed from the pathology could have been the cause of the fatal arrhythmia although they disagreed as to the likelihood of it being the only cause. All experts agreed that the pathology did not reveal any unstable or "hot" plaque in Mr. Demarco's arteries. All experts agreed that large transmural or replacement scarring on the heart muscle is not required to trigger a fatal arrhythmia. All experts agreed that a 50% occlusion of an artery is considered insignificant.

[219] I accept the pathology evidence as interpreted by Drs. Butany and Chaisson as follows:

- i) Mr. DeMarco died of a fatal arrhythmia;
- ii) Mr. DeMarco did not suffer a heart attack;
- iii) Mr. DeMarco had microscopic/interstitial scarring on his heart muscle;
- iv) there was no evidence of unstable plaque in the two stenosed arteries sampled;
- v) there was no evidence of a thrombosis (blood clot) in the two stenosed arteries sampled;
- vi) Mr. DeMarco suffered from two-vessel coronary artery disease with narrowing of 70 to 80%;
- vii) scarring is irreversible and cannot be treated;
- viii) vasospasm occurs in arteries less-stenosed; and
- ix) Mr. DeMarco's fatal arrhythmia developed when he was asleep or at rest with low oxygen demands.

[220] The defendant submits that had Mr. DeMarco received treatment of his coronary artery disease prior to August 20, 2011, he would have remained at the same risk of suffering a

fatal arrhythmia on August 20, 2011 which ultimately lead to his death. It was myocardial scarring that caused the fatal arrhythmia and not his untreated coronary artery disease.

- [221] The plaintiffs' sole expert on the cause of Mr. DeMarco's fatal arrhythmia was Dr. Butany. Dr. Butany was of the opinion that it was unlikely Mr. DeMarco's fatal arrhythmia was caused solely by his interstitial, microscopic, patchy fibrosis. Dr. Butany was of the opinion that an ischemic event, evidence of which was not detectable on autopsy, either alone or together with the microscopic scarring revealed on the autopsy, caused Mr. DeMarco's fatal arrhythmia.
- [222] Dr. Butany opined the cause of the fatal arrhythmia was a coronary vasospasm. A coronary vasospasm occurs during periods of high oxygen demand. The undisputed evidence is that Mr. DeMarco suffered a fatal arrhythmia while sleeping or at rest and while in a state of low oxygen demand.
- [223] Dr. Butany is an esteemed, highly-regarded, cardiac pathologist. While I accept that, generally speaking, it is unlikely microscopic scarring, on its own, will cause a fatal arrhythmia, Mr. DeMarco's pathology establishes that in his case that unlikely event (microscopic scarring causing a fatal arrhythmia) most likely occurred. The analysis would be different had there been evidence of unstable plaque in Mr. DeMarco's narrowed arteries or evidence of a thrombosis or evidence of a heart attack. However, without evidence of any of the other possible causes of sudden cardiac death, those other possible causes or combinations of causes are speculative, at best. An inference cannot be drawn that any of these possibilities is the likely mechanism of death. At its highest, Dr. Butany's opinion that a vasospasm caused Mr. DeMarco's sudden cardiac death is a possibility as opposed to a likelihood. His evidence as to the cause of Mr. DeMarco's sudden death cannot satisfy the plaintiffs' burden of proof.
- [224] In *Kozak v. Funk*, the Saskatchewan Court of Appeal held that an inference of causation must be based on an appropriate evidentiary foundation:

There must however be an evidentiary base on which the factual determination can be made. In my opinion, the trial judge erred when he concluded that "an inference may be drawn where common sense dictates, despite the lack of evidence or medical unanimity regarding factual causation." While the issue of causation can frequently be determined by the use of common sense, there must be an evidentiary underpinning on which to base the finding of fact which includes all the circumstances surrounding the accident or accidents and the medical evidence...Snell does not stand for the proposition that an inference may be drawn in the absence of evidence or that the court is entitled to bridge, by way of inference, a gap in the evidence.²⁹ [Emphasis Added.]

²⁹ *Kozak v. Funk*, [1997] S.J. No. 743 (C.A.), at para. 22, BOA Tab 15.

- [225] I am not permitted to draw an inference that coronary artery disease was the cause of Mr. DeMarco's fatal arrhythmia simply because Mr. DeMarco had coronary artery disease.
- [226] The evidence establishes that Mr. DeMarco had interstitial patchy fibrosis in both ventricles spread throughout the walls. All experts agree that scar tissue in and of itself in the absence of any of the other possible mechanisms of death can cause a fatal arrhythmia. There is no evidence, rather only speculation by Dr. Butany, that an ischemic event such as a spasm or thrombosis (which leaves no residue observable post-mortem) occurred and that the undetectable ischemic event, alone, or in combination with the scarring, caused the fatal arrhythmia.
- [227] I accept that it is *possible* the mechanism of Mr. DeMarco's fatal arrhythmia was a vasospasm (Option B) or a combination of a vasospasm and scarring (Options B and C); however, the only pathological evidence of the existence of any of the four options is scarring making Option C, on its own, the most likely cause of Mr. DeMarco's fatal arrhythmia. Based on the pathological evidence, Option C is the most probable mechanism with Option B, or a combination of Options B and C, mere possibilities. The plaintiffs have not met the burden of proof on the issue of causation.
- [228] The only inference which can be drawn from the evidence is that, on a balance of probabilities, the scarring present on Mr. DeMarco's heart was the mechanism or cause of Mr. DeMarco's fatal arrhythmia. Dr. Butany's evidence rises only to the level of possibility not probability and therefore does not rise to the level necessary to prove that "but for" the failure to treat Mr. DeMarco's coronary artery disease, Mr. DeMarco would not have suffered a fatal arrhythmia on August 20, 2011.

DAMAGES

A. *FLA* Claims

- [229] The parties have agreed to the assessment of the plaintiffs' damage claims for the loss of care, guidance, and companionship arising from Mr. DeMarco's death pursuant to s. 61 of the *FLA*, as follows:
- (a) Lisa - \$100,000
 - (b) Emily - \$100,000
 - (c) Franco - \$35,000
 - (d) Julia - \$35,000
 - (e) Gabriel - \$35,000
 - (f) Rita - \$15,000
 - (g) Frank - \$15,000.

The parties further agree that prejudgment interest accrues on the foregoing amounts at the rate of 5% being the applicable rate for non-pecuniary losses provided for in rule 53.10 of the *Rules of Civil Procedure*.

- [230] Emily is under a disability and the court must therefore decide whether the agreement reached between the parties is in her best interests. The report of Dr. Warren Blume described Emily's physical and cognitive impairments. She has a chronic seizure disorder requiring medication. She suffers from scoliosis, needs assistance with personal hygiene, is unable to articulate her feelings, cannot drive and has poor concentration. I find that Emily is severely cognitively disabled and is unable to live independently.
- [231] At the time of her father's death, Emily was entirely dependent on her father and mother for care. Lisa testified that she intends to continue to reside with and care for Emily for as long as she is able. Emily's impact statement was read to the court. It is apparent that Emily shared a close bond with her father and they spent a considerable amount of social time together. Given her dependence on her father, I find Mr. DeMarco's death had a profound impact on Emily akin to the loss of a spouse. The amount agreed by the parties is similar to the amount awarded the disabled daughter of the deceased in *Wilcox v. "Miss Megan" (The)*³⁰. As here, the amount approved by the court for the disabled child was identical to the amount awarded to the spouse in *Wilcox*.
- [232] I hereby approve the amount agreed by the parties on account of Emily's *FLA* claim in the amount of \$100,000.

B. Special Damages

- [233] The parties have agreed to past special damages (for the 6-year period commencing September 2012) claimed by Lisa totaling \$50,169.08 comprised of the following amounts:
- (a) funeral expenses: \$32,556.76;
 - (b) yard maintenance expenses: \$10,691.40; and
 - (c) home maintenance expenses: \$6,920.92.
- [234] The parties agree that the funeral expenses accrue interest at the rate of 1.3% as provided for in section 128 of the *Courts of Justice Act* and have agreed that the yard and maintenance expenses accrue interest at 0.65% (or 1/2 x 1.3%).
- [235] Lisa also claims for future yard and home maintenance expenses. Using the average annual amount for these expenses of \$2,935.38 per year, the cumulative present value of the total amount claimed to Mr. DeMarco's retirement at age 67 for these future expenses is \$22,241.37.

³⁰ 2008 FC 506.

- [236] Lisa is also claiming past expenses for this same 6-year period relating to amounts paid to Hugh Elliott, the estimator hired by CHC, in the amount of \$53,972.21 and amounts paid to the site-supervisor, Barry Balcom, in the amount of \$33,148.55. Lisa claims these services would have been provided by Mr. DeMarco.
- [237] The plaintiffs also claims for future annual expenses of amounts paid to the estimator and site supervisor to Mr. DeMarco's retirement at age 67 years using an average yearly cost of \$14,520. The present value of these amounts is \$110,018.04.
- [238] I find that Lisa is entitled to damages for future yard and home maintenance expenses in the amount claimed totaling \$22,241.37. This finding is merely an extension of the agreement reached by the parties with respect to entitlement to damages for past yard and home maintenance expenses.
- [239] I find that the plaintiffs are not entitled to damages for past and future expenses relating to amounts paid or to be paid, as the case may be, to the estimator and site supervisor. These are not expenses of the plaintiffs. These are expenses of CHC. CHC is not a party to this action.

C. Economic Loss – Dependency Loss Claims of Lisa and Emily

(i) Evidence

(a) Lisa DeMarco

- [240] Lisa testified that Carlo DeMarco completed four years of a drafting course at Ryerson Polytechnical Institute however did not graduate. The couple was married in 1984. Mr. DeMarco worked at various jobs in the construction industry before starting CHC in May 1993. Lisa worked at Bell Canada as a clerk. She took a buyout from Bell before joining CHC in 1997. CHC consisted of Carlo and Lisa. CHC has no employees rather it contracted with various sub trades to complete projects. Lisa was responsible for the administrative work relating to CHC including scheduling meetings and preparing cheques and invoices. Lisa split her time between her administrative duties and caring for the home and the children. She recalls working long hours each day and 6 to 7 days per week.
- [241] In 1997 CHC purchased 3 - 4 lots and also purchased the lot on Chantry Place upon which the family home was built in 1997. The business office for CHC was located in the family residence. Ultimately this home was sold in 2015 and Lisa, Emily and Frank moved to their current home located on Carriage Hill Drive. This home was also built by Mr. DeMarco. CHC continues to be run out of the family home.
- [242] In the early years of CHC, its business involved building smaller homes. The market plummeted in 2008 and 2009 and CHC diversified into commercial and home renovations. At the time of trial, the focus of the business was on renovations. CHC has not built any new homes since 2011. Lisa testified that she believes if Mr. DeMarco had

lived he would have continued to build new homes. At all times, CHC maintained a license to build homes and continues to carry that license, reapplying every year.

- [243] Lisa estimated that CHC built more than 250 new homes during the period 1993 – 2011. Mr. DeMarco struggled to find funding to purchase new lots. Lisa described the industry as competitive and at the time of Mr. DeMarco's death, no bank was prepared to loan money to CHC to purchase more lots. However, Mr. DeMarco was able to find financing through high interest private lenders and lines of credit provided by family members.
- [244] At the time of Mr. DeMarco's death, CHC owned one lot and it was intended that Mr. DeMarco would design and build a high-end home on this lot. Lisa admitted during cross-examination that the last time CHC had purchased any building lots was in 2004/05 – 7 years prior to Mr. DeMarco's death. By 2011, CHC was focussing on residential and commercial renovations. Lisa maintains that CHC was weathering the economic storm and intended to return to new home building once the economy improved.
- [245] After Mr. DeMarco's death, CHC hired an individual to estimate on commercial jobs to assist Lisa in submitting bids. Another individual was hired to supervise the projects and the subtrades. Lisa believed that once the economy improved, CHC would return to building homes exclusively and that renovations were just a temporary business focus.
- [246] Lisa has a grade 12 education. She is not dating and has not remarried. As she has always done, Lisa continues to care for Emily including assisting with her personal hygiene, preparing meals and socializing with her. She testified that she is not prepared to leave CHC and go back into the workforce because continuing to run CHC from the family home allows her to continue to care for Emily at home.

(b) Emily

- [247] Emily suffers from epilepsy. She has reduced cognitive functioning. She has a grade 12 certificate. She currently receives ODSP. Lisa testified that Emily will never live independently. Emily is under the care of several doctors.

(c) Gabriel

- [248] Gabriel testified that he began helping out his father with construction projects when he was approximately nine years of age. He was working full-time at CHC until his father's death in August 2011. At that time, Gabriel was 21 years of age. Gabriel testified that at the time of his father's passing the major focus of the business of CHC was new residential construction.

- [249] Gabriel assisted as a general labourer helping to organize trades, cleaning jobsites and framing. During cross-examination he admitted his father similarly performed manual labour. Gabriel corroborated the evidence of Lisa. He described his mother's role as administrative in nature. She split her time between working as a secretary for CHC and taking care of the home and in particular caring for Emily.
- [250] Outside maintenance of the home was the responsibility of Mr. DeMarco, Gabriel, and Franco. Lisa was primarily responsible for looking after Emily. Gabriel recalled discussing with his father that they would continue to build new homes together. Gabriel described building new homes as his father's passion. Gabriel testified that if his father was still alive they would both be working at CHC building new homes.

(d) Ted Melchers

- [251] Ted Melchers was called as a lay-witness in support of the plaintiffs' claim for damages.
- [252] Mr. Melchers is the president of Melchers Construction Limited, a home building business operating in London for the past 32 years. The business performs a wide variety of construction activities including new home building, commercial building and commercial and residential renovations in and around the city of London.
- [253] Mr. Melchers was familiar with CHC; however, neither Carlo DeMarco nor Lisa are known to him. He recalls that CHC was a company which built higher-end homes.
- [254] Mr. Melchers employs construction managers in his business - an inside construction manager also known as a residential project manager and an outside construction manager.
- [255] The inside project manager organizes the trades for the new homes, meets with the client to go over aspects of the construction and follows through with the trades to ensure that the project is completed on time. An inside project manager earns approximately \$75,000 per year.
- [256] The outside manager takes care of the day-to-day sub- trades and work at the specific site. The outside construction manager earns approximately \$70,000 per year.
- [257] A draftsman that works with Melchers earns approximately 90,000 - \$100,000 per year. The draftsman is involved with designing new homes. A person with four years drafting education and 25 years' experience building and designing home would be qualified to work as an inside residential project manager and, possibly, as a draftsman.
- [258] Mr. Melchers testified that his business with regards to new home builds has been the busiest it has been in the past two years.
- [259] During cross-examination Mr. Melchers confirmed that his business employs approximately 30 people and has an office staff performing administrative functions for the business. He further confirmed that the draftsman was someone who worked in the

office full time with construction managers working at the project site or out of the office with the customers.

- [260] Mr. Melchers confirmed that, in addition to home building, his business involved constructing medical clinics, car dealerships, warehouses and at the time of giving testimony was constructing a condo community. In addition, Melchers was constructing 12 new homes and the outside construction manager was responsible for overseeing construction of these homes.

(ii) Historical Income of Lisa and Carlo DeMarco

- [261] Based on the income tax returns for Lisa and Mr. DeMarco for the period 2006 – 2011, the average yearly income generated for the DeMarco family was \$97,118.66. This average income dropped to \$56,000 if the income generated for the DeMarco family in 2005 is included.

(iii) Expert Evidence

(a) James Hoare

- [262] Mr. Hoare testified as an expert in support of the plaintiffs' claims for damages. Mr. Hoare is both a chartered accountant and a chartered business valuator specializing in valuing businesses and providing litigation support services. Mr. Hoare has also been certified as a specialist in investigative and forensic accounting by the Canadian Institute of Chartered Accountants. Forensic accounting is an investigation of accounting records and transactions including damage quantification in wrongful death cases.
- [263] Mr. Hoare was qualified as an expert in forensic accounting and qualified to provide expert opinion evidence with respect to the assessment of dependency claims in wrongful death cases.
- [264] Mr. Hoare prepared four reports – a report dated May 22, 2015, a report dated February 16, 2017, a report dated August 17, 2017 and a report dated June 6, 2018.
- [265] Mr. Hoare explained that a loss of dependency in a wrongful death action relates to the income generated by the deceased which otherwise would have been available to the surviving family members. It is a loss to the surviving spouse and the dependents because income that otherwise would have been generated by the deceased is no longer available to his dependents. From gross income, income tax and other statutory deductions are deducted together with a percentage of the income relating to the deceased's individual needs for food, clothing and other necessities of life, to arrive at a net dependency loss.
- [266] Mr. Hoare gave as an example a two income household in which both spouses earn income. In that scenario, it is assumed that 40% of the net income (after tax) would be consumed by the deceased, leaving 60% available to the family unit. Dependent children

are then allocated 4% each to a maximum of four dependent children. In a single-income family unit, 70% is allocated to the family unit and 30% to the deceased.

- [267] For purposes of his opinion, Mr. Hoare assumed the DeMarco family unit was a single-income household because CHC was the sole source of income for the family unit. He also allocated 10% (rather than 4%) relative to Emily because she was special needs and would be dependent on Emily's care for the balance of her life. In addition, he assumed increased costs relative to her care. Mr. Hoare assumed additional expenses relating to Emily for medications, transportation costs, clothing and incidental expenses.
- [268] Mr. Hoare calculated the dependency loss to age 67. In selecting age 67, Mr. Hoare referred to statistics indicating that self-employed individuals work longer than their employed counterparts. In addition, Mr. DeMarco had not made recent contributions to an RRSP and did not have a pension to look to upon retirement suggesting he would need to continue to work from a financial perspective.
- [269] In arriving at his calculation, Mr. Hoare assumed that Mr. DeMarco and Lisa were equal shareholders in CHC with Mr. DeMarco responsible for the day-to-day operations of the company and Lisa responsible for part-time administrative duties for the company with her primary duties relating to the care of Emily and the household, generally.
- [270] Mr. Hoare noted that the company continued to operate after Mr. DeMarco's death. However, the company lost contracts and incurred operating losses in 2012 and 2013. Mr. Hoare picked a valuation date of September 30, 2015 for purposes of his initial calculations. In preparing his opinion as to dependency losses, Mr. Hoare referred to Mr. DeMarco's income tax returns for the period 2006 through 2011 together with his terminal return. He also reviewed unaudited financial statements for the company from 2005 through 2014 and CHC's corporate tax return for the year ending September 30, 2013.
- [271] In preparing his opinion, Mr. Hoare was asked to assume that CHC carried on business as a new home builder and renovator. In addition, he took into account that CHC had built a number of new homes in the years prior to Mr. DeMarco's death; however, had struggled in the years immediately prior to his death with new home-builds and so had expanded its business into home renovations.
- [272] Mr. Hoare assumed that had Mr. DeMarco not died, CHC would have emphasized its renovation business in the short term because the real estate market/new home market in the London area had declined significantly due to the recession in 2008 and 2009.
- [273] Mr. Hoare relied on housing market information for the London market released in the fall of 2016 by the Canadian Mortgage and Housing Corporation. This information indicated an increase in the housing market and in particular an increase in housing starts for single-detached units in London. The reports reviewed by Mr. Hoare showed little growth in the housing start-ups to 2015; however, a significant increase in the number of housing starts in 2016 and 2017. Mr. Hoare assumed CHC would return to building

houses in the same manner as it did in 2006 and 2007 and that revenue would increase relative to the increase in the number of housing starts.

- [274] Mr. Hoare reviewed a job bank website for the Government of Canada with respect to the wages of construction managers in the London area. From this information, Mr. Hoare confirmed that the median wage for construction managers in London Ontario was \$36.06 per hour with a low-end of \$24.04 and a high-end of \$57.69 per hour. Having regard to his education and experience, Mr. Hoare attributed an hourly wage of \$36.06 to Mr. DeMarco. Once extrapolated, this hourly wage generates an annual income of \$75,000 based on Mr. DeMarco working 2000 hours per annum. This is consistent with the evidence of Mr. Melchers.
- [275] From the financial records reviewed, Mr. Hoare confirmed that Mr. DeMarco had 25 years' experience building new homes and built approximately 15 new homes during that time. Mr. Hoare considered three scenarios (applying dependency factors of 60/4% and 70/10% and adjusting for joint mortality and present day value) in calculating Lisa's and Emily's past loss of dependency claim for the period October 1, 2011 to September 10, 2018 and future loss of dependency claims (to Mr. DeMarco's date of retirement at age 67 years).
- [276] Under Scenario 1, Mr. Hoare calculated past and future dependency loss based on Mr. DeMarco's income relying upon historic information of CHC. Scenario 1 does not take into consideration any inflationary changes or changes in the housing market subsequent to 2015. Mr. Hoare took into account the average gross sales for the 7 years prior to Mr. DeMarco's death less the cost of sales and fixed expenses and the contribution of Lisa (estimated to be \$8-\$10,000 per annum) to arrive at a range of projected net income for Mr. DeMarco of \$38,134-\$50,394. Using a 60/4% dependency factor the range of past dependency loss is \$165,700- \$219,300. Using a 70/10% dependency factor the range is \$207,100-\$274,200. Using a 60/4% dependency factor the range of future dependency loss was calculated as \$185,500-\$245,200. Using a 70/10% dependency factor the range is \$232,000-\$306,700.
- [277] Under Scenario 2 Mr. Hoare calculated past and future dependency loss based on Mr. DeMarco's income relying upon the same historic information of CHC relied upon in Scenario 1; however, Mr. Hoare applied a growth factor to Mr. DeMarco's average net income relative to the increase in the number of housing starts in London in 2016 and 2017. Using a 60/4% dependency factor the range of past dependency loss is \$257,400-\$318,500. Using a 70/10% dependency factor the range is \$321,800-\$398,100. Using a 60/4% dependency factor the range of future dependency loss was calculated as \$476,200-\$560,200. Using a 70/10% dependency factor the range is \$595,900-\$700,900.
- [278] Under Scenario 3 Mr. Hoare calculated dependency loss assuming Mr. DeMarco was earning \$75,000 per annum working as a construction manager. Using a 60/4% dependency factor the past dependency loss was calculated as \$237,300. Using a 70/10% dependency factor the past dependency loss was calculated as \$296,600. Using a 60/4% dependency factor the future dependency loss was calculated as \$277,600. Using a 70/10% dependency factor the future dependency loss was calculated as \$347,500.

- [279] Mr. Hoare estimated Mr. DeMarco's retirement income assuming he received Old Age Security in the amount of \$8,167 per annum and Canadian Pension Plan benefits in the amount of \$8,472 per annum from the age of 67 years to his estimated date of death. Mr. Hoare assumed Mr. DeMarco would live to age 83 years based on normal life expectancy tables.
- [280] Mr. Hoare then calculated the joint cumulative mortality adjusted present value of the future loss of dependency of Lisa and Emily during Mr. DeMarco's retirement years to the age of 83. Using a 60/4% dependency factor the dependency loss was calculated as \$104,600. Using a 70/10% dependency factor the dependency loss was calculated as \$131,700.

(b) Frank Lekivetz

- [281] Mr. Lekivetz holds a degree in Economics and a Masters in Business Administration. His career has focussed on quantification of damages in personal injury cases. Unlike Mr. Hoare, Mr. Lekivetz is not a chartered business valuator nor certified as an investigative forensic accountant. He is not an accountant.
- [282] Mr. Lekivetz was qualified to opine on the loss of financial support experienced by Lisa and Emily as a result of the death of Mr. DeMarco in August 2011. He admitted to having no expertise in business valuation. His evidence was restricted to his opinion on the loss of support for Lisa and Emily.
- [283] Mr. Lekivetz was asked to provide an opinion regarding the loss of financial support experienced by Lisa and Emily as a result of the death of Mr. DeMarco. He calculated Mr. DeMarco's earning capacity for the balance of his life at three retirement ages 61, age 63 and age 65 years. Mr. Lekivetz calculated dependency loss using the approach utilized by Mr. Hoare in his Scenario 3. However, Mr. Lekivetz did not agree Mr. DeMarco would have been employed as a construction manager. Rather, he calculated DeMarco's income earning capacity based on the average employment income of a home building and renovations manager in the London Ontario area working full time. Using Statistics Canada information, the average annual employment earnings of a home building and renovations manager in 2010 was \$52,027.
- [284] Mr. Lekivetz was of the view that a home building and renovation manager was an occupation most closely aligned to Mr. DeMarco's general job tasks at CHC. Based on the Statistics Canada information a home-building and renovation manager is responsible for planning the building of a new home, planning renovations, consulting with clientele, dealing with subcontractors and trades and various other activities. In calculating dependency loss, Mr. Lekivetz used the average earnings of a home building and renovations manager from the 2011 Canada National Household Survey and adjusted for inflation for each year from 2010 to 2018 to arrive at taxable income of \$59,304. He then deducted tax payable on Mr. DeMarco's income and adjusted for mortality to arrive at the amount of income available for distribution for use by the entire family. That amount was \$41,344.

- [285] Like Mr. Hoare, Mr. Lekivetz assumed Emily would remain dependent on her father for financial support for the balance of her life; however, he did not accept a dependency rate of 10% because there was no evidence of Emily's increased care costs. Instead, he applied a 4% dependency rate for Emily. He was further of the opinion that the appropriate dependency rate for Lisa was 60% (rather than 70%) because both Mr. Demarco and Lisa were income-earners at the time of Mr. Demarco's death.
- [286] After adjusting for mortality and applying a dependency rate of 60/4% Mr. Lekivetz calculated a total past loss of support for the period August 21, 2011 through September 10, 2018 of \$183,100 (\$171,700 for Lisa and \$11,400 for Emily). He was asked to calculate the total future dependency loss for each of Lisa and Emily assuming Mr. DeMarco retired at age 61, 63 and 65 years. After adjusting for mortality and applying a dependency rate of 60/4% Mr. Lekivetz calculated a total future loss of support assuming Mr. DeMarco retired at age 65 and assuming a life expectancy of 71 years of age, of \$150,600 (\$137,600 for Lisa and \$9,500 for Emily).³¹

(c) Dr. Fitchett

- [287] With respect to Mr. DeMarco's life expectancy, if he had not suffered cardiac arrest on August 21, 2011 and had received medical and surgical interventions prior to August 20, 2011, Dr. Fitchett's opinion was that Mr. DeMarco had a life expectancy of approximately 15 years being approximately 10 years shorter than a man of his age. That is, Mr. DeMarco had a less than 50% chance of being alive in 15 years. During cross-examination, Dr. Fitchett amended his opinion to accord with Mr. DeMarco's age at the time of his death and increased his life expectancy to 20 years having regard to the absence of risk factors.

(iv) Factual Findings

- [288] Mr. Hoare based his estimate of Mr. DeMarco's income earning capacity on the historical performance of CHC. He also considered Mr. DeMarco's income earning capacity based on the revenue of CHC increasing in accordance with the housing starts in London Ontario. Last Mr. Hoare considered Mr. DeMarco's income earning capacity based on the average earnings of a construction manager in the London Ontario area.
- [289] Mr. Lekivetz was of the view that the historical earnings of CHC were not a reliable indicator of Mr. DeMarco's future earning capacity because at the time of his death the company was having difficulty obtaining financing for future housing projects and had ceased, at least temporarily, constructing new homes and was focussing its business on commercial and home renovations.
- [290] During closing submissions, plaintiffs' counsel asked that I accept Mr. Hoare's calculations under Scenario 3 when calculating Emily and Lisa's dependency loss and

³¹Damages were increased by \$3,500 to correct Schedule 2-2 of Exhibit No. 57 to reflect an assumption that no tax was payable on CPP and OAS.

that I disregard the approach taken under Scenarios 1 and 2. Scenario 3 is most closely aligned with the approach taken by Mr. Lekivetz.

- [291] Based on the plaintiffs' revised position on the issue of damages, I am not required to consider whether the assumptions underlying Scenarios 1 Scenario 2 are supported by the evidence. I need only consider whether the assumptions underlying Scenario 3 are supported by the evidence.
- [292] Under Scenario 3, the central issue is whether an income consistent with the median hourly wage for a construction manager as Mr. Hoare suggests or an income consistent with the median hourly wage for a home building and renovations manager as suggested by Mr. Lekivetz, should be attributed to Mr. DeMarco.
- [293] Mr. DeMarco had 4 years of post-secondary education. He had 25 years' experience designing and building homes in the London area. In his capacity as owner of CHC, he performed many roles with respect to each construction project and new home builds including overseeing the management and construction of each new home. In fact, he built and designed the family home on Chantry Place. Photographs of this home filed as exhibits at trial reflect the extremely high quality Mr. DeMarco's work. I find his skill-set far exceeded that of a home building and renovations manager and that Mr. DeMarco had the income earning capacity most closely aligned with that of a construction manager.
- [294] Lisa's role in the household was a traditional one. However, I find that during the years preceding Mr. DeMarco's death, she did perform administrative duties for CHC and those administrative duties had a value. I do not accept that the value of her services was on par with the value of services provided by Mr. DeMarco to CHC. In addition to performing administrative tasks, Lisa was primarily responsible for caring for Emily and maintaining the household. I estimate the value of Lisa's contributions to CHC prior to Mr. DeMarco's death to be equivalent to a part-time income. Mr. DeMarco was primarily responsible for maintaining the lifestyle of the DeMarco family.
- [295] I accept that the appropriate dependency rate to apply to the loss of dependency income of Lisa is 70% of Mr. DeMarco's after-tax income earning capacity. While I have found that Lisa was a part-time income earner; her income was tied to CHC. CHC was the sole income source for the DeMarco family unit. It is therefore appropriate to use a dependency rate for Lisa of 70% rather than 60%.
- [296] I accept that the appropriate dependency rate for Emily is 10%. To support a 10% dependency rate, the plaintiffs are not required to provide evidence of additional future care costs. The plaintiff must only establish that it is unlikely Emily will ever attain independence. Unlike the average dependent child who typically gains independence between the ages of 18 and 22 years, I find Emily will remain a dependent for the balance of her life thereby justifying an increased dependency rate of 10%.
- [297] Being self-employed, I accept, as was assumed by Mr. Hoare, that Mr. DeMarco likely would have continued working to age 67. However, Mr. Hoare assumed a normal life

expectancy of 83 years and did not reduce Mr. DeMarco's life expectancy having regard to his advanced coronary artery disease. I accept Dr. Fitchett's revised opinion as to Mr. DeMarco's life expectancy and find that Mr. DeMarco had a life expectancy of 71 years of age.

- [298] I accept Mr. Lekivetz' estimate of Mr. DeMarco's CPP benefits at age 67 of \$10,731. I note this estimate of retirement income is more favourable than the estimate provided by the plaintiffs' expert. I further accept Mr. Lekivetz' estimate of Mr. DeMarco's annual Old Age Pension benefit as \$7,160 for total annual retirement income of \$17,891.

(v) Calculation of Dependency Loss Claims of Lisa and Emily

- [299] I calculate the total past dependency loss claim of Lisa and Emily for the period October 1, 2011 to September 10, 2018 using a 70/10% dependency factor as established by Mr. Hoare in Scenario 3, to be \$296,600.
- [300] I calculate the total future dependency loss claim of Lisa and Emily for the period September 10, 2018 to Mr. DeMarco's retirement at age 67 years using a 70/10% dependency factor as established by Mr. Hoare in Scenario 3, to be \$347,500.
- [301] I find that the total mortality adjusted present value of the future dependency loss claim of Lisa and Emily from the date of retirement at 67 years to Mr. DeMarco's expected date of death at 71 years, is that amount calculated using a 70/10% dependency factor on annual retirement income of \$17,891. After applying the applicable cumulative present value factor listed in the Appendix A and Appendix B to Exhibit #26, for Lisa and Emily, respectively, this amount is \$135,519 for Lisa and \$19,908 for Emily.

D. PJI

- [302] As noted above, the parties agreed that the appropriate pre-judgment interest rates were as follows: 5% on non-pecuniary losses, 1.3% on funeral expenses and 0.65% on yard and home maintenance expenses.
- [303] I find that the prejudgment interest rate applicable to Emily and Lisa's past dependency losses is 0.65% (1/2 of 1.3%).

DISPOSITION

- [304] Action dismissed.

COSTS

[305] In the event the parties cannot agree on costs, I will receive written submissions, not exceeding 5 pages in length (exclusive of case law, a Costs Outline and Bill of Costs) from the defendant within 30 days; and, from the plaintiffs within 15 days thereafter. Reply submissions, if any, within 5 days thereafter.

“Justice A.K. Mitchell”
Justice A. K. Mitchell

Released: May 3, 2019

CITATION: The Estate of Carlo DeMarco et al. v. Dr. Martin et al., 2019 ONSC 2788

COURT FILE NO.: 1539-13

DATE: 2019/05/03

ONTARIO

SUPERIOR COURT OF JUSTICE

BETWEEN:

The Estate of Carlo DeMarco

-and-

Dr. Rene Frances Martin

REASONS FOR JUDGMENT

Justice A.K. Mitchell

Released: May 3, 2019